



硕士学位论文

(学术学位)



资本市场自由化是否推动数字化转型？

来自中国 A 股纳入 MSCI 新兴市场指数的证据

Does Capital Market Liberalization Drive Digital Transformation?

Evidence from Chinese A-Shares'

Inclusion in the MSCI Emerging Markets Index

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摘要

新兴经济体资本市场自由化水平不足不利于企业融资，进而制约其转型发展。本文以中国 A 股分阶段纳入 MSCI 新兴市场指数为外生制度冲击，基于 2010-2022 年间 2,712 家上市公司数据，借助双重差分识别策略，并结合事件研究和倾向得分匹配等方法，研究新兴市场经济体的资本市场自由化能否推动企业数字化转型。

实证结果显示，在 TWFE 基准设定下，MSCI 纳入对应的数字化转型系数为 0.39 个对数点（约 48%）。但在分期处理与队列 ATT 等现代 DID 诊断中，平均效应明显收敛且统计精度下降，表明传统 TWFE 在异质处理效应下可能高估总体效应。事件研究结果显示，处理前不存在系统性趋势差异，处理后效应呈累积特征。安慰剂检验、替代标准误聚类、受限样本与替代被解释变量检验在方向上支持主结论。异质性分析发现，效果主要集中于高技术行业和制度环境较好的地区，说明吸收能力与制度互补性是关键约束条件。

机制检验显示，融资渠道证据具有“结构调整强、综合约束弱”的特征：银行杠杆和现金流指标显著改善，但 KZ 指数不显著；治理渠道指标方向一致且显著，但需考虑合规性与被动持股稀释等制度解释。知识溢出渠道因关键中介变量未完全并入样本，暂不作直接系数结论。总体而言，国际金融一体化对企业数字化升级具有条件性促进作用，但效应大小依赖估计方法、测度设定与企业实施能力。政策含义上，资本市场开放改革需要与信息披露质量提升、公司治理改进和区域数字基础条件协同推进，才能更稳定地转化为企业持续数字化投入。对企业而言，纳入带来的融资机会与外部监督约束并存，长期收益仍取决于组织执行能力与技术吸收能力。

关键词：资本市场自由化；数字化转型；MSCI 纳入；中国 A 股；新兴市场；计算文本分析；双重差分

Abstract

The insufficient level of capital market liberalization in emerging economies hinders corporate financing, thus restricting their transformation and development. This paper takes China's phased inclusion in the MSCI Emerging Markets Index as an exogenous institutional shock. Based on data from 2,712 listed companies between 2010 and 2022, and using a difference-in-differences (DID) identification strategy, along with event studies and propensity score matching, this study investigates whether capital market liberalization in emerging market economies can promote corporate digital transformation.

Empirical results show that in the TWFE benchmark specification, MSCI inclusion is associated with a 0.39 log-point increase in digital transformation intensity (about 48%). However, modern staggered DID diagnostics based on cohort-specific ATT estimates are materially smaller and less precise, indicating substantial estimator sensitivity under heterogeneous treatment timing. Event-study evidence supports no systematic pre-treatment divergence and shows cumulative post-treatment dynamics. Placebo tests, alternative clustering strategies, restricted-sample checks, and alternative outcomes provide directional robustness.

Heterogeneity results indicate stronger effects in high-technology sectors and institutionally developed regions, consistent with an absorptive-capacity interpretation. Mechanism evidence is nuanced: financing-channel results are more consistent with financing-structure adjustment than broad constraint relaxation (bank leverage and cash-flow proxies are significant, while the KZ index is not), governance proxies are directionally strong but institutionally interpretable with compliance and passive-ownership caveats, and direct knowledge-spillover estimation remains deferred due to mediator-data limitations. Overall, international financial integration appears to be a conditional contributor to digital upgrading in emerging markets, with effect magnitudes dependent on estimator choice, measurement assumptions, and firm-level implementation capacity. From a policy perspective, market-opening reforms should be coordinated with stronger disclosure quality, governance discipline, and regional digital support conditions to translate external capital exposure into sustained firm-level digital investment.

For firms, inclusion-related financing opportunities and external monitoring pressures coexist, and long-run gains depend on implementation and absorptive capacity rather than inclusion status alone.

Keywords: Capital market liberalization; Digital transformation; MSCI inclusion; Chinese A-shares; Emerging markets; Computational text analysis; difference-in-differences

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List of Abbreviations

Abbreviation	Full Term
AI	Artificial Intelligence
ATT	Average Treatment Effect on the Treated
C1	Basic Manufacturing (CSRC Subcategory Code)
C2	Equipment Manufacturing (CSRC Subcategory Code)
C3	High-End Manufacturing (CSRC Subcategory Code)
C27	Transportation Equipment Industry (CSRC Industry Code)
C38	Electrical Equipment Industry (CSRC Industry Code)
C39	Instrumentation Industry (CSRC Industry Code)
CF	Cash Flow
CNINFO	China Securities Information Disclosure Platform
CNRDS	China Research Data Service
CNY	Chinese Yuan
CSMAR	China Stock Market and Accounting Research Database
CSRC	China Securities Regulatory Commission
DID	difference-in-differences
DOI	Digital Object Identifier
DT	Digital Transformation
DT_Comp	Composite Digital Transformation Index
EM	Emerging Markets
ERP	Enterprise Resource Planning
FE	Fixed Effects
GTA	GTA Information Technology Co.
H1	Hypothesis 1
H2	Hypothesis 2
H3	Hypothesis 3
HHI	Herfindahl-Hirschman Index
HMM	Hidden Markov Model
IaaS	Infrastructure as a Service
ICT	Information and Communication Technology
IMF	International Monetary Fund

List of Abbreviations

Abbreviation	Full Term
IoT	Internet of Things
IP	Intellectual Property
IT	Information Technology
KZ Index	Kaplan-Zingales Financing Constraint Index
MSCI	Morgan Stanley Capital International
NLP	Natural Language Processing
NPV	Net Present Value
PaaS	Platform as a Service
PSM	propensity score matching
PSM-DID	propensity score matching difference-in-differences
QFII	Qualified Foreign Institutional Investor
R ²	Coefficient of Determination
RMB	Renminbi
ROA	Return on Assets
ROE	Return on Equity
RQFII	RMB Qualified Foreign Institutional Investor
SaaS	Software as a Service
SD	Standard Deviation
SE	Standard Error
SIC	Standard Industrial Classification
SME	Small and Medium-Sized Enterprise
STAR	Science and Technology Innovation Board
TWFE	two-way fixed effects

Chapter 1 Introduction

This study examines whether capital market liberalization is associated with changes in firm-level digital transformation by exploiting the phased inclusion of Chinese A-shares in the MSCI Emerging Markets Index as quasi-exogenous policy variation. It models digital transformation as a long-horizon intangible investment process shaped by managerial capability and external financing and governance conditions. The empirical analysis therefore provides identification-oriented evidence on whether financial opening relaxes these constraints and increases transformation intensity, rather than only describing whether firms transform.

1.1 Research Background

The relationship between capital market liberalization and corporate digital transformation sits at the intersection of development economics, corporate finance, and technology management. Over the last two decades, many emerging economies have gradually opened domestic capital markets to international investors while firms have faced growing pressure to digitize production, operations, and governance. Yet these two developments have mostly been studied in isolation. Liberalization research tends to focus on valuation, liquidity, and investment efficiency, while the digital transformation literature emphasizes internal capabilities and technology readiness^[1]. A key question at their intersection remains open: can international capital-market integration raise firm-level digital transformation, and if so, through which channels?

The starting point of this study is that digital transformation is not only a technological or managerial process, but also a financing and governance process shaped by market institutions. In many emerging markets, digital programs require high fixed costs, involve uncertain returns, unfold over long horizons, and depend on organizational learning complementarities^[2]. Even when expected social gains are high, these features can create financing frictions that delay or discourage adoption^[3]. Capital market liberalization can shift these investment conditions by broadening the investor base, lowering financing costs, and strengthening governance discipline^[4]. Understanding this link is

therefore important for both scholarship and policy.

The Chinese background is especially important for this research. China features scale, diversity, and institutional growth in ways that make it ideal for empirical study^[5]. It is one of the world's largest stock markets and a leading hub for industrial digital transformation, yet capital account liberalization has mostly been gradual and driven by policy^[6]. This slow process results in notable policy events with predictable timing. Among these, the inclusion of Chinese A-share firms in the MSCI Emerging Markets Index stands out as a key externally imposed reform that increased firms' exposure to international investors.

The practical relevance of this study is equally significant. Policymakers increasingly treat digital transformation as a core source of productivity growth and competitiveness. However, firms differ substantially in their capacity to implement large-scale transformation programs^[7]. If access to global capital and investor networks shapes adoption, market-opening reforms can generate broader technological spillovers than is usually assumed^[8]. At the same time, liberalization may widen regional and firm-level capability gaps if benefits accrue mainly to firms that already hold structural advantages. This tension underscores the need for rigorous empirical evaluation.

Measurement considerations constitute an additional key motivation. Much of the existing evidence relies on managerial surveys or narrow proxies such as IT spending and patent counts. These indicators often suffer from inconsistent coverage, self-reporting bias, and weak comparability across time^[9]. To address this issue, the study constructs a firm-year digital-transformation index from annual-report disclosures using computational text analysis. Combining this high-coverage measurement strategy with a quasi-experimental design allows the analysis to provide evidence that is both identification-oriented and broad in empirical scope.

Chen et al.^[10] show that increased analyst coverage and improved information accuracy are two mechanisms through which digital transformation strengthens firms' information environments. Cheng and Masron^[11] further argue that regulatory uncertainty can intensify firms' digital responses, highlighting the central role of external institutional conditions in shaping transformation behavior. Du and Wang^[12] find that

the innovation benefits of digital transformation are significantly amplified by financial support, particularly through capital markets, with effects significant at the 1% level. Zhang and Wang^[13] report that digitally transformed firms face lower equity financing costs, whereas Xue and Zhang^[14] document improvements in debt-financing outcomes and overall firm performance. Taken together, this evidence demonstrates that the financial consequences of digital transformation are conditional on financing environment, disclosure quality, and governance structure.

This study builds on prior work while moving beyond correlation-based inference. Earlier studies often focus on one channel or one outcome and do not fully address treatment endogeneity. By contrast, this analysis combines mechanism-oriented evidence, a long panel (2010–2022), and a quasi-experimental policy shock to assess whether liberalization plausibly contributes to digital transformation intensity under maintained identification assumptions. Conceptually, digital transformation is treated as cumulative strategic investment: firms adjust governance, reassess financing constraints, and update expectations over time. Liberalization is expected to influence this path through at least three channels: financing-constraint relaxation, governance and disclosure upgrading through international oversight, and expanded access to cross-border knowledge networks. These channels anchor the empirical framework.

Three main motivations follow from these observations. First, there is a significant open question about the relationship between firm-level technological upgrading and global finance. Second, methodologically rigorous evidence from a major emerging economy remains scarce. Third, firms and regulators seeking to align capital-market reform with digital-development objectives need micro-level evidence to guide policy design.

1.2 Research Questions and Objectives

Building on these motivations, the study addresses three research questions.

- First, does inclusion in the MSCI Emerging Markets Index, a proxy for capital market liberalization, significantly increase firm-level digital transformation intensity? This is the study’s primary identification question. Specifically, the analysis eval-

uates whether treated firms exhibit a sustained post-inclusion increase in digital transformation relative to otherwise comparable untreated firms.

- Second, do treatment effects vary by technology domain, industry, region, and firm characteristics? Digital transformation is not a single technology, but a bundle of investments with distinct capital-intensity and capability requirements. The study tests whether effects are stronger in domains such as AI, big data, and cloud computing; whether responses differ across technology-intensive and traditional industries; and whether regional institutional quality shapes outcomes.
- Third, how does liberalization affect digital transformation? The study investigates three channels using theory and prior evidence: funding restrictions, governance and monitoring, and knowledge spillovers. In empirical implementation, the analysis directly tests the first two channels and treats the knowledge channel as theory-grounded but exploratory in the current sample.

Methodologically, the study aims to integrate validated NLP-based measurement with quasi-experimental econometrics in a transparent and reproducible large-panel framework. Substantively, it evaluates whether international financial integration can serve as a catalyst for firm-level technological upgrading. From a policy perspective, it provides evidence relevant to regulators, institutional investors, and corporate decision-makers seeking to understand the technology consequences of market-opening reforms.

To achieve these objectives, the analysis proceeds in five steps: generating high-coverage firm-year digital transformation metrics from annual reports; estimating treatment effects using baseline and dynamic DID techniques; evaluating inferential stability through robustness and sensitivity tests; identifying boundary conditions through heterogeneity analysis; and assessing financing, governance, and knowledge channels through mechanism regressions.

1.3 The Empirical Setting: China's A-Share Market and MSCI Index Inclusion

The empirical background for this argument is the gradual inclusion of Chinese A-shares in the MSCI Emerging Markets Index, a watershed moment in China's financial opening process^[15]. Prior to 2018, global investors had limited access to China's domestic equity market due to ownership restrictions, quota systems, and market-access frictions. Although conduits such as QFII, RQFII, and the Stock Connect programs existed, broad passive global allocation to mainland A-shares remained limited^[16].

MSCI inclusion changed this environment by connecting eligible A-share companies to benchmark-driven international capital flows^[17]. MSCI indexes are often used by global institutional investors and index-tracking funds. Being included in the MSCI Emerging Markets Index affects active portfolio building, analyst focus, and investor relations practices, in addition to passive investment flows^[6]. The reform therefore modified firms' exposure to external financing and oversight conditions in a way that is economically significant.

The phased inclusion process is particularly useful for identification. Phase I was implemented in June 2018 with 234 large-cap A-shares at a 5% inclusion factor following the June 2017 announcement. By November 2019, Phase II expanded coverage to 472 securities with a higher inclusion factor. This phased implementation provides clear treatment timing and observable pre- and post-policy windows. A key advantage of this setting is that inclusion criteria were largely rule-based and externally determined. Eligibility was based on market capitalization, liquidity, and Stock Connect accessibility rather than on discretionary judgments about firms' digital strategies^[18]. Although no real-world setting is fully random, this institutional design strengthens the plausibility that inclusion captures exogenous variation in international capital access relative to firms' underlying digital transformation trajectories.

China is also an ideal context because firms operate under substantial heterogeneity in regional development, industry composition, ownership structures, and governance quality. This heterogeneity allows the study to test whether liberalization effects are

uniform or conditional on absorptive capacity^[19]. Zhang et al.^[20] show that treatment effects vary systematically with regional market development, ownership structure, and firm size, with stronger responses in certain non-Eastern and non-state-firm contexts. Embedding these moderation patterns within the MSCI setting helps move beyond average treatment effects toward a richer understanding of distributional outcomes.

1.4 Overview of Research Design and Data

The empirical design combines identification-oriented quasi-experimental inference with large-scale computational measurement. The baseline model is a two-way fixed-effects DID that compares firms included in MSCI (treated) with firms not included (control) before and after inclusion. This baseline is complemented by event-study estimation for dynamic effects and pre-trend validation, propensity-score-matched DID for improved overlap on observables, and placebo-based falsification tests.

The data cover 35,264 firm-year observations from 2010 to 2022, representing 2,712 Chinese listed firms after sample cleaning. Core financial and governance variables are drawn from CSMAR, while treatment status is derived from MSCI constituent lists. Digital transformation outcomes are constructed from annual-report text using NLP-based keyword extraction across six technology domains: AI/machine learning, big data, cloud computing, IoT, blockchain/distributed ledger technologies, and enterprise digitization programs.

The baseline outcome is a composite index using log transformation of aggregated keyword frequencies. This approach addresses right skewness and enhances interpretability when disclosure dispersion is high. For robustness, alternative outcomes are also used, including domain-specific indices, normalized measures adjusted for report length, and additional transformations to assess sensitivity to scaling choices.

Mechanism analysis is embedded in the design rather than added as a descriptive task. Proxies related to financing constraints and governance quality are estimated through structured regressions to check whether channel patterns match the conceptual framework. The knowledge-spillover channel is retained as a theory-grounded mecha-

nism and interpreted using exploratory evidence in the current sample.

Overall, the design is intentionally triangulated: one institutional shock, multiple estimators, multiple outcome definitions, and multiple mechanism checks. This triangulation increases confidence that the results are not artifacts of any single model choice, while still requiring cautious interpretation under estimator sensitivity and measurement limits.

1.5 Significance of Research

This study contributes to three strands of literature: capital market liberalization, digital transformation economics, and computational empirical methods.

- First, it expands the liberalization literature beyond traditional market outcomes. Previous research often focuses on liquidity, valuation, return co-movement, or financing costs. This study provides evidence that liberalization also affects firms' technological paths, connecting financial integration to long-term productivity-related behavior.
- Second, it advances the digital transformation literature by highlighting external financial environments as a key factor in corporate digital strategy. While much existing research focuses on internal capabilities, managerial cognition, and organizational change, the findings here indicate that access to international capital and investor oversight can significantly influence the pace and direction of transformation.
- Third, it offers mechanism-level evidence that directly tests financing and governance channels under a common treatment design, while treating knowledge spillovers as a theory-grounded exploratory channel with suggestive interpretation.
- Fourth, the study enhances methodology by combining quasi-experimental inference with validated Chinese-language text analytics at scale.
- Fifth, it contributes to the emerging-market comparative literature by offering high-resolution evidence from China.
- Sixth, it provides policy implications based on micro-level evidence. Regulators in-

interested in the broader development impact of capital market reform can use these findings to assess whether market-opening policies promote strategic upgrading.

1.6 Thesis Structure

The thesis is organized in seven chapters.

Chapter 1 introduces the research problem, outlines the motivation and objectives, describes the empirical setting, and previews the expected contributions.

Chapter 2 reviews the theoretical and empirical literature on capital market liberalization, digital transformation, and text-based measurement, identifying the gaps that motivate this study.

Chapter 3 provides the institutional background, including China's pre-liberalization capital market structure and the MSCI inclusion process. It also develops the formal hypotheses.

Chapter 4 presents Research Design I, covering sample construction, variable definitions, model specification, and the causal identification strategy.

Chapter 5 presents Research Design II, documenting the computational text-analysis framework used to measure digital transformation from annual-report disclosures.

Chapter 6 reports the empirical results—baseline estimates, event-study dynamics, robustness tests, heterogeneity analysis, and mechanism evidence—followed by a discussion of the findings.

Chapter 7 concludes with a synthesis of contributions, limitations, and directions for future research.

Chapter 2 Literature Review

This chapter positions the thesis within the literature on capital market liberalization, digital transformation, and computational measurement. It emphasizes where existing evidence is descriptive or partial and where identification and mechanism integration remain incomplete. These gaps motivate the empirical strategy implemented in the subsequent chapters.

2.1 Capital Market Liberalization: Theory and Evidence

2.1.1 Theoretical Foundations of Financial Liberalization

Capital market liberalization generally refers to removing barriers that separate domestic financial systems from international portfolio flows^[21]. Opening equity markets can, in principle, improve allocative efficiency by channeling capital toward more productive firms, reducing financing costs through broader investor participation, and improving risk sharing through more diversified ownership^[22]. The standard welfare argument predicts stronger gains for firms facing tighter financing constraints^[23]. In segmented markets, limited domestic savings, home bias, and information frictions keep external finance expensive, and liberalization can reduce these distortions by intensifying investor competition and deepening market liquidity^[24].

Theoretical work also highlights important non-linearities. The benefits of liberalization depend on institutional quality, legal enforcement, disclosure standards, and market infrastructure^[25]. Where institutions are weak, liberalization may amplify volatility or redirect resources toward short-horizon opportunities rather than long-horizon productive investment^[26]. At the firm level, liberalization changes the decision environment facing managers and controlling shareholders^[11]: it affects the cost of external finance, investor horizons, and the intensity of external monitoring^[27].

The MSCI inclusion mechanism is particularly informative because it combines signaling and flow effects. Inclusion can automatically trigger passive fund allocations while simultaneously signaling market quality and investability to active global investors.

2.1.2 Effects on Corporate Investment, Innovation, and Governance

Empirical studies generally find that market opening influences corporate behavior through investment and governance channels^[28]. On the investment side, reduced financing frictions can broaden firms' feasible project options^[29]. Firms typically increase R&D, capital expenditure, and strategic intangible investments when expected financing constraints lessen^[23]. On the governance side, foreign institutional investors can strengthen monitoring and raise transparency expectations.

Evidence from prior work supports this logic. Chen et al.^[10] show that digital transformation is associated with higher analyst coverage and improved information accuracy. Du and Wang^[12] provide direct evidence that financial support strengthens the innovation effects of digital transformation, particularly in environments with tighter conventional financing constraints. Chen and Zhang^[3] and Feng et al.^[30] extend this argument to firms' financing structures.

2.1.3 Evidence from Emerging Market Contexts

Emerging markets provide valuable evidence because they combine high growth potential with institutional frictions that highlight financing channels. Studies in these markets often detect stronger real effects of financial reforms compared to more developed markets where constraints are already less severe^[31]. The Chinese evidence is especially useful due to detailed data and specific policy information^[32]. However, a direct causal link from liberalization shocks to firm-level digital transformation outcomes is less developed.

Zhang et al.^[20] show that contextual heterogeneity is central: estimated effects vary with regional market development, ownership structure, and firm size. This pattern implies that pooled average treatment estimates may mask economically meaningful distributional differences. Stiglitz^[33] similarly emphasizes that policy shocks interact with organizational absorptive capacity.

2.2 Digital Transformation: Conceptual Framework and Determinants

2.2.1 Defining Digital Transformation: Technological and Organizational Dimensions

Digital transformation is more than technology acquisition. It involves an ongoing, firm-wide reconfiguration of value creation, operations, and organizational coordination through digital technologies^[34]. A narrow input-based view that focuses only on hardware and software spending misses the strategic and organizational conditions that determine whether digital investment translates into productivity gains. Conceptually, the process has at least three layers^[14,35]: infrastructure adoption, process transformation, and organizational transformation.

These layers depend on each other. Infrastructure without process integration yields limited returns^[36]. Process changes without organizational incentives can stall. Organizational commitment without funding remains superficial^[14]. Digital transformation is therefore best understood as a coordinated investment journey rather than a single technological event^[2].

2.2.2 Core Technology Domains: AI, Big Data, Cloud Computing, IoT, and Blockchain

The technology areas highlighted in this study align with key trends in modern corporate digital transformation. Artificial intelligence and machine learning support prediction, optimization, and automation across both front-end and back-end operations^[37]. Big data systems enable companies to turn large, fast-moving information into operational insights. Cloud computing offers scalable infrastructure that reduces fixed ICT costs^[38]. Internet of Things technologies link physical assets to digital monitoring and control systems^[39]. Blockchain and distributed ledger technology provide traceability, tamper resistance, and coordination across organizational boundaries^[40]. Enterprise digitization initiatives combine these technologies with ERP upgrades, process engineering, and platform strategies.

Empirically, these domains are related but not interchangeable margins of transformation. AI/ML and advanced analytics are usually data- and talent-intensive, with higher setup costs and longer learning curves. Cloud adoption often lowers entry barriers and accelerates deployment, while IoT initiatives require coordinated redesign of software and physical production systems. Blockchain applications are often narrower, but can be highly relevant in traceability-intensive supply chains. As a result, measured digital intensity can move at different speeds across domains even under similar financing conditions^[41-42].

This heterogeneity implies that aggregate digital indices should be interpreted as averages over technologies with different risk profiles, adjustment speeds, and implementation bottlenecks, rather than as a single homogeneous investment decision. That interpretation is important for later empirical sections, where treatment effects are estimated on both composite and technology-specific outcomes.

2.2.3 Drivers, Barriers, and Firm-Level Heterogeneity

Drivers of digital transformation include financial slack, managerial capability, data infrastructure, and complementary human capital. Recent studies also emphasize organizational execution quality: cross-functional coordination, digital leadership commitment, and internal data-governance standards all increase the probability that projects scale beyond pilot stages^[43].

Barriers are equally important. High uncertainty about payoffs can delay investment, especially for firms under pressure to meet short-horizon performance targets. Talent shortages in data engineering, AI deployment, and cybersecurity create bottlenecks that raise implementation risk^[43]. Legacy IT architectures and fragmented databases further increase integration costs and extend rollout cycles.

These frictions imply heterogeneous response capacities. Larger firms may capture stronger scale economies in platform investment and digital talent recruitment^[44]. Firms in economically developed regions may access deeper supplier ecosystems and larger pools of specialized labor. By contrast, resource-constrained firms may prioritize modular adoption and postpone system-wide upgrades.

Ownership and market structure also matter. Non-state firms often face tighter product-market discipline, which can strengthen incentives to adopt productivity-enhancing digital strategies^[45]. Overall, the literature reveals that average effects can conceal substantial cross-sectional dispersion. This motivates explicit heterogeneity analysis by region, industry, and firm characteristics in the empirical chapter.

2.2.4 Digital Transformation in Emerging Markets: Opportunities and Constraints

Emerging markets offer a digital transformation landscape with both growth opportunities and structural constraints^[46]. In these settings, access to external capital markets can play a catalytic role. International investors may provide financing and governance pressure. They may also transmit benchmarking standards and stronger long-horizon capability expectations^[47].

At the same time, external financing does not mechanically translate into successful digital upgrading. The pass-through from capital access to implementation depends on absorptive capacity, including managerial quality, organizational readiness, and local complementary infrastructure. This mechanism helps explain why treatment effects differ materially across firms and regions in emerging economies.

China is especially informative because advanced digital ecosystems coexist with substantial regional and firm-level inequality in capabilities and financing conditions^[48]. This combination provides a strong context for identifying not only whether liberalization affects digital transformation, but also where and for whom the effect is strongest.

2.3 Mechanisms Linking Capital Market Liberalization to Digital Transformation

2.3.1 The Financing Constraint Channel

The financing-constraint channel is the most direct link between liberalization and digital transformation. Digital programs often require substantial upfront spending on software systems, data architecture, cloud migration, machine-learning platforms, and

organizational training^[49]. By broadening access to equity investors and lowering required returns, liberalization can reduce the user cost of capital for these investments^[50].

Existing studies provide direct support for this channel. Hordofa^[51] demonstrates that capital-market financial support significantly strengthens the innovation effects of digital transformation, while Liu^[15] documents lower equity financing costs among firms undergoing digital upgrading. Together, these findings suggest that financing and digital transformation are mutually reinforcing. The channel is unlikely to be homogeneous across firms: firms with initially binding financing constraints should exhibit larger responses^[52].

2.3.2 Corporate Governance and International Investor Monitoring

A second mechanism is governance upgrading through international investor monitoring. Foreign institutional investors often demand higher disclosure quality, stronger board accountability, and clearer strategic communication^[27]. Under these conditions, managers face stronger incentives to prioritize long-term value-enhancing projects and reduce agency-driven underinvestment.

Ma et al.^[53] provide supporting evidence that digital transformation is associated with improved information environments through increased analyst coverage. Li^[54] further establishes that governance effects can interact with ownership structure.

2.3.3 Knowledge Spillovers and Global Best Practice Diffusion

A third mechanism operates through information and knowledge spillovers. Capital market liberalization boosts interactions between domestic firms and global investors, analysts, and intermediaries^[8,22]. Through investor meetings, analyst reports, benchmarking exercises, and governance engagement, firms can access implementation templates and performance standards^[55]. This mechanism aligns with broader themes in the literature^[24,56-57]. Evidence from competitive manufacturing environments further suggests that digital-technology spillovers can be strategically state-dependent across rival firms rather than mechanically uniform^[58].

2.3.4 Market Signaling, Certification, and Reputational Effects

Liberalization can affect digital transformation through signaling and certification processes. Inclusion in a major global benchmark is often interpreted as recognition of minimum standards in accessibility, liquidity, and market relevance^[54]. This signal can lower external uncertainty about firm quality. Dong^[19] further distinguishes substantive transformation from symbolic signaling, an important distinction when interpreting post-inclusion disclosure behavior.

2.4 Text-Based Measurement of Firm Characteristics

2.4.1 Computational Approaches in Corporate Finance and Economics

The rise of text-as-data methods has expanded the scope of empirical corporate research by enabling high-frequency, large-sample analysis^[59]. For studying digital transformation, computational text methods are particularly valuable because direct accounting measures are often incomplete^[60]. These approaches improve scalability and consistency compared to manual coding^[61], though they require careful validation^[62].

2.4.2 NLP-Based Measurement of Innovation and Digital Activity

Afzal^[63] confirms that Chinese-language annual reports can provide credible NLP-based indicators of digital activity. Kumar^[64] further argues for domain-cluster keyword extraction rather than generic buzzword counts. Li et al.^[65] position NLP-based measurement as a complement to, rather than a substitute for, financial indicators.

2.4.3 Validation Strategies for Text-Derived Indices

Validation is central to any text-based empirical strategy. This study applies three validation layers: content validity, construct validity, and robustness validity. Content validity is established through careful dictionary design and expert review. Construct validity is evaluated by comparing text-derived indices with related observable variables. Robustness validity is assessed through alternative outcome definitions and algorithmic specifications.

2.4.4 Challenges in Chinese-Language Disclosure Environments

Chinese-language textual analysis presents technical and interpretive challenges^[9,65]. Chinese text segmentation is complex because characters are written without spaces. Another challenge is variation in strategic disclosure^[59]. A third challenge involves distinguishing between intention and implementation.

Despite these challenges, Chinese annual reports remain some of the most comprehensive and standardized data sources for long-term firm-level analysis. With careful design and validation, they offer a strong empirical foundation for studying digital transformation at scale.

2.5 Research Gaps and Positioning of This Study

The literature review reveals four core gaps that motivate this study.

- **Linkage gap:** Research on capital market liberalization and research on digital transformation have mostly developed independently.
- **Identification gap:** Many studies rely on cross-sectional correlations or panel models that face endogeneity concerns.
- **Mechanism integration gap:** Existing evidence suggests financing, governance, and knowledge channels are all plausible but often examined in isolation.
- **Measurement scalability gap:** Direct digital investment metrics are often incomplete in emerging-market accounting systems.

This study addresses these gaps by treating MSCI A-share inclusion as a quasi-natural experiment to identify the effect of liberalization on digital transformation under explicit identification assumptions. While prior work has examined individual pieces of this relationship, no existing study integrates these elements within a single coherent design. The thesis is structured to provide that integrated identification-oriented framework.

Chapter 3 translates this framework into testable hypotheses and institutional identification conditions.

Chapter 3 Institutional Background and Hypothesis Development

This chapter provides the institutional context for the empirical analysis and develops the formal hypotheses. It begins with the pre-liberalization structure of China's A-share market and the role of Stock Connect as implementation infrastructure for foreign participation. It then describes the MSCI inclusion timeline and rule-based eligibility criteria, explaining why the reform generates plausibly exogenous variation in international capital access. The chapter concludes by translating the conceptual framework from Chapter 2 into testable, falsifiable hypotheses for the empirical analysis.

3.1 China's Capital Market Structure Prior to Liberalization

3.1.1 The A-Share Market and Domestic Investor Segmentation

China's A-share market—the renminbi-denominated equity market on the Shanghai and Shenzhen exchanges—was historically governed by strict foreign ownership restrictions and capital controls, creating near-complete segmentation from international capital flows^[66]. Before liberalization, foreign investors could access mainland equities mainly through QFII (Qualified Foreign Institutional Investor) and RQFII channels, both constrained by quotas and administrative barriers. Under this segmented regime, listed firms relied primarily on domestic capital, typically associated with shorter investment horizons, weaker governance pressure, and limited exposure to global best practices.

Figure 3-1 reports pre-treatment segmentation patterns for 2010–2017. The chart shows standardized mean gaps between firms eventually included in MSCI (2018–2022) and firms never included. The gaps are economically large and statistically precise for firm size (1.20 standard deviations, $p < 0.01$), total market value (0.80 standard deviations, $p < 0.01$), and independent-director representation (0.47 standard deviations, $p < 0.01$). This pattern confirms that the subset of firms later connected to global capital was already structurally distinct within the domestic A-share universe.

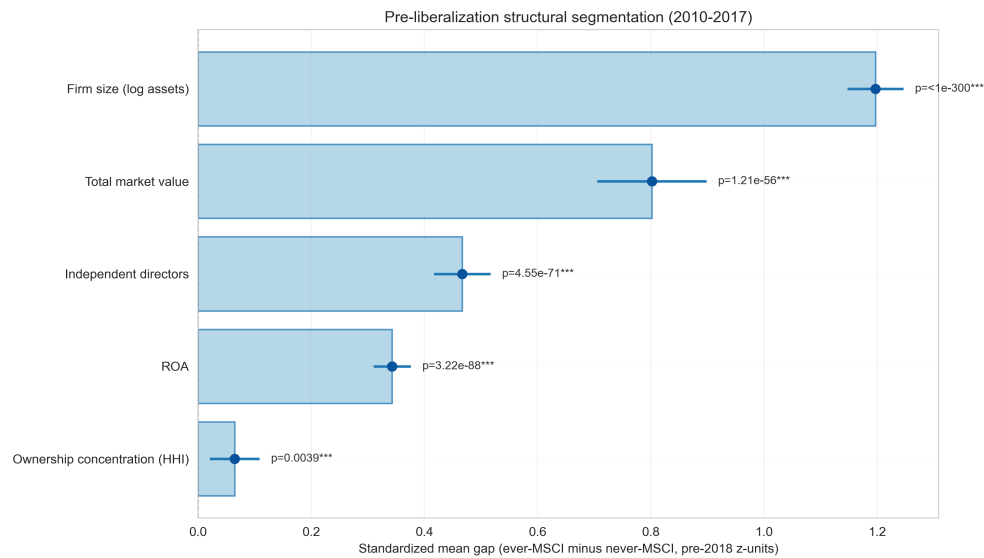


Figure 3-1 Data Profile: Pre-Liberalization Structural Segmentation (2010–2017)

3.1.2 Stock Connect Programs as the Technical Infrastructure for Liberalization

The technical foundation for liberalization was built through Stock Connect: Shanghai–Hong Kong Stock Connect (November 2014) and Shenzhen–Hong Kong Stock Connect (December 2016)^[67]. These programs established quota systems, currency-conversion arrangements, and settlement protocols that allowed foreign participation in A-shares while preserving capital-control constraints. By 2018, the platform had matured enough to accommodate the larger portfolio reallocations associated with MSCI inclusion^[23].

3.2 The MSCI Emerging Markets Index and the A-Share Inclusion Process

3.2.1 MSCI as a Global Benchmark: Scale, Influence, and Index Composition

The MSCI Emerging Markets Index is a primary benchmark for global institutional investment in developing-market equities. In 2018, roughly \$1.9 trillion in assets tracked or benchmarked MSCI emerging-market indices. Because passive funds mechanically follow index composition and active managers often benchmark against it,

index membership materially affects capital allocation toward included firms^[17].

3.2.2 Inclusion Criteria: Market Capitalization, Liquidity, and Stock Connect Accessibility

MSCI's inclusion criteria were mechanical rather than discretionary: (i) a minimum free-float adjusted market capitalization threshold; (ii) minimum liquidity requirements based on annualized trading volume and turnover ratios; and (iii) accessibility through Stock Connect programs. Importantly, these criteria are independent of firms' underlying digital transformation paths or performance, offering a plausible source of exogenous variation in international capital access.

To make this point data-based, Figure 3-2 estimates a 2017 cross-sectional logit model where the dependent variable is eventual MSCI inclusion. The odds-ratio estimates show strong and statistically significant associations for each rule-based eligibility proxy: log total market value (OR = 17.04, $p < 0.01$), log free-float market value (OR = 2.55, $p < 0.01$), and liquidity proxy (OR = 3.90, $p < 0.01$). This pattern supports the interpretation that inclusion was driven by measurable market-scale and tradability conditions rather than discretionary selection on digital strategy.

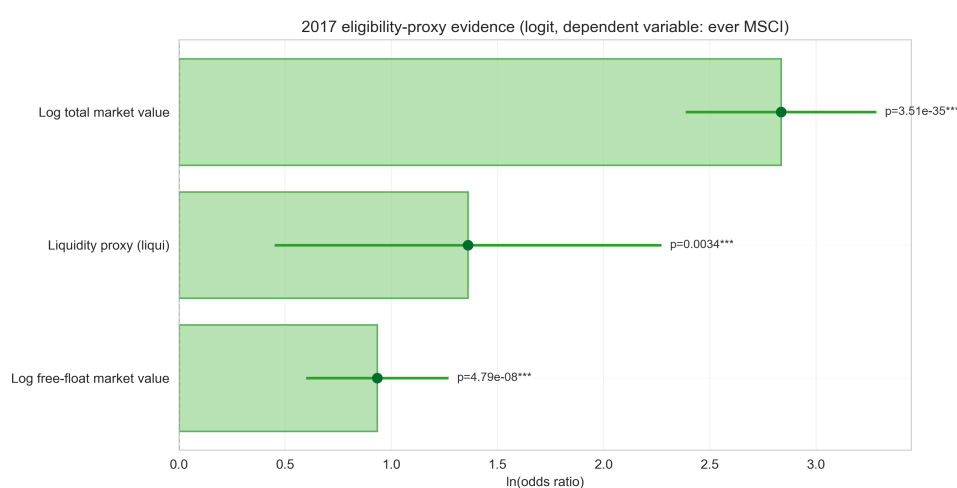


Figure 3-2 Data Profile: Eligibility-Proxy Evidence from 2017 Logit Model

3.2.3 Pre-Announcement and Phased Implementation Timeline

MSCI announced the inclusion decision on June 20, 2017—a full year before the first implementation tranche—providing firms with time to prepare for international in-

vestor engagement. The announcement itself constitutes a clean event that allows testing of announcement effects separately from implementation effects^[68].

3.2.4 Phase I (June 2018): 234 Large-Cap A-Shares at 5% Inclusion Factor

The initial tranche, launched on June 1, 2018, incorporated 234 large-cap A-shares at a 5% inclusion factor. This accounted for about 0.73% of the total MSCI Emerging Markets Index. The modest size aimed to evaluate market preparedness and reduce impact, while allowing the included companies to experience international passive fund inflows for the first time.

3.2.5 Phase II (November 2019): Expansion to 472 Securities at 20% Factor

By November 2019, the index grew to include 472 A-shares across large, mid, and small-cap categories, with a 20% inclusion rate. This significantly boosted the presence of Chinese stocks in the index and broadened the scope to include smaller firms, expanding the universe of treated firms.

Figure 3-3 visualizes the realized treatment rollout in the matched panel used for empirical analysis. The number of MSCI firms jumps from zero before 2018 to 151 in 2018 and 374 in 2019, then stabilizes around 372–373 during 2020–2022. In share terms, MSCI firms increase from 4.8% of listed firms in 2018 to 11.4% in 2019 (95% confidence interval approximately [10.3%, 12.5%]), consistent with a sharp, policy-timed treatment expansion.

3.2.6 Quasi-Experimental Properties of the Inclusion Design

MSCI inclusion has three features that make it a strong quasi-experimental setting. First, eligibility criteria are mechanical and not based on firms' digital-transformation strategies. Second, the rollout timeline was set externally by MSCI rather than by Chinese regulators or individual firms. Third, phased implementation creates clear pre- and post-treatment windows with multiple cohorts, which supports staggered-adoption analysis.

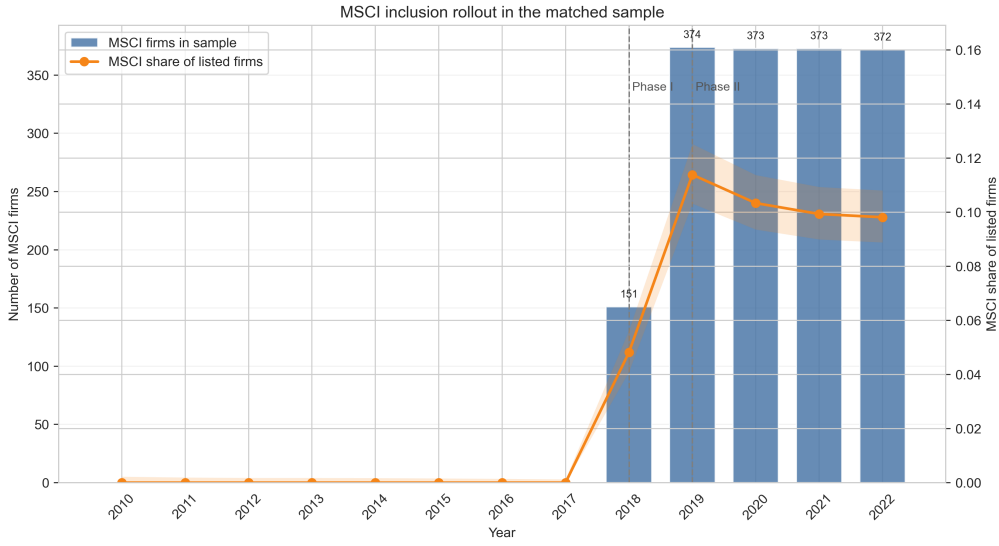


Figure 3-3 Data Profile: MSCI Inclusion Rollout in the Matched Sample

3.3 Hypothesis Development

Building on the institutional evidence above, this section develops hypotheses in three parts: the direct effect (Section 3.3.1), transmission mechanisms (Section 3.3.2), and heterogeneity (Section 3.3.3). The data profiles in Sections 3.1 and 3.2 provide the empirical grounding for these hypotheses.

For clarity, we define the treatment shock as $X_{it} = \text{MSCI}_i \times \text{Post}_t$ and the outcome as $Y_{it} = \text{DT}_{it}$ (firm-level digital transformation intensity). The core identification question is how X affects Y , and through which channels this effect is transmitted.

To formalize channel and moderation logic in advance, let M_{it}^k denote channel-specific mediators ($k \in \{\text{Fin}, \text{Gov}, \text{Know}\}$) and let H_{it} denote heterogeneous conversion conditions (regional institutions, industry technology intensity, and firm absorptive capacity). The hypotheses below are developed under three criteria: (i) each claim is linked to an explicit economic mechanism; (ii) each mechanism implies observable, directional predictions; and (iii) each prediction is potentially falsifiable by coefficient signs, dynamic patterns, or interaction tests. This structure keeps hypothesis statements tightly aligned with the empirical identification strategy.

3.3.1 Main Effect: Direct Impact of Capital Market Liberalization on Digital Transformation

The data profiles in Section 3.1 and Section 3.2 motivate the main-effect hypothesis from two complementary facts: (i) the treatment rollout is sharply time-delimited to the post-2018 window (Figure 3-3); and (ii) inclusion is strongly tied to mechanical market-scale and liquidity criteria (Figure 3-2). Together, these features support interpreting MSCI inclusion as an external capital-access shock suitable for DID-style identification testing rather than discretionary selection on firms' digital strategy.

From an intertemporal corporate-finance perspective, firms choose digital investment by comparing expected discounted benefits against financing and implementation costs. A reduced-form decision representation is given by Equation (3-1):

$$D_{it}^* = \arg \max_D \{ \mathbb{E}[\Pi_{it+1}(D)] - C(D; r_{it}, \phi_{it}) \} \quad (3-1)$$

where r_{it} is the effective external financing cost and ϕ_{it} captures governance and implementation frictions. Liberalization can reduce segmentation-related frictions, improve liquidity, and broaden investor participation. These changes jointly lower the user cost of capital and expand the feasible digital project set^[21-24]. In addition, greater exposure to international investors can tighten strategic monitoring and reduce agency-related underinvestment in long-horizon intangibles^[11,25,27].

Because digital transformation requires high fixed costs, complementary organizational change, and uncertain payoffs, it is especially sensitive to financing conditions and governance quality^[14,34-35]. Under standard comparative statics, if $\partial D^*/\partial r < 0$ and $\partial D^*/\partial \phi < 0$, and MSCI inclusion decreases both r and ϕ , then the expected sign is $\partial D^*/\partial X > 0$. This prediction is theory-consistent but not deterministic: weak absorptive capacity or short-horizon managerial incentives can dampen pass-through in some firms, which is why average and heterogeneous effects are separately hypothesized.

Hypothesis 1 (H1 — Main Effect): Capital market liberalization, proxied by MSCI inclusion, increases firm-level digital transformation intensity.

Direct treatment path: $X \rightarrow Y$.

The empirical signature of H1 has three parts. First, in two-way fixed-effects DID models, the coefficient on X_{it} should be positive after conditioning on common shocks and time-invariant firm heterogeneity. Second, in event-study models, pre-treatment coefficients should remain near zero, while post-treatment coefficients should move upward if the identifying assumptions are plausible. Third, under a capability-building interpretation, effects should persist or accumulate over time rather than appearing as a one-period spike.

Empirical implication: In DID and event-study models, the treatment interaction should be positive, pre-trend coefficients should be statistically indistinguishable from zero, and post-treatment dynamic coefficients should remain above pre-treatment coefficients.

3.3.2 Impact Mechanism: Three Transmission Channels from Liberalization to Digital Transformation

The mechanism analysis distinguishes three transmission channels: financing-constraint relaxation, governance improvement, and knowledge spillovers. The structural differences shown in Figure 3-1 and the rule-based eligibility profile in Figure 3-2 indicate that capital-access and monitoring channels are empirically plausible in this setting. Chapter 6 reports formal channel regressions for financing and governance and treats knowledge-channel evidence as exploratory in the current sample.

Conceptually, the average treatment effect can be decomposed as Equation (3-2):

$$\frac{\partial Y}{\partial X} = \sum_{k \in \{\text{Fin, Gov, Know}\}} \frac{\partial Y}{\partial M^k} \frac{\partial M^k}{\partial X} + \frac{\partial Y}{\partial X} \Big|_{\text{direct}} \quad (3-2)$$

This decomposition is useful because it separates total reduced-form effects from channel-specific evidence and clarifies what can be tested with currently available mediator data.

Hypothesis 2 (H2 — Impact Mechanism): The positive effect of capital market liberalization on digital transformation operates through three channels: (i) financing-constraint relaxation, (ii) governance improvement via external monitoring, and (iii) knowledge-spillover effects from global investor and information networks.

H2a (Financing channel): $X \rightarrow M^{\text{Fin}} \rightarrow Y$. In segmented markets, external-finance premia and liquidity frictions constrain long-horizon intangible investment. Liberalization should reduce these frictions by broadening the investor base and improving capital-market depth^[15,49-51]. The prediction is directionally clear: measures of financing tightness should decline and internal funding flexibility should improve after inclusion, with stronger effects expected among ex-ante constrained firms^[52].

H2b (Governance channel): $X \rightarrow M^{\text{Gov}} \rightarrow Y$. International investors can increase monitoring intensity, disclosure discipline, and board accountability, reducing agency frictions that otherwise bias managers toward short-horizon or low-transparency projects^[27,53-54]. The testable prediction is that governance-quality proxies improve after inclusion and that these improvements are associated with larger digital-transformation responses.

H2c (Knowledge-spillover channel): $X \rightarrow M^{\text{Know}} \rightarrow Y$. Liberalization can also expand information links to global analysts, institutions, and benchmarking networks, reducing learning costs and uncertainty in technology adoption^[8,22,55-57]. This channel is theoretically central but empirically demanding because it requires direct mediator series (for example, analyst coverage and foreign ownership) to avoid weak proxy interpretation^[58].

This framework leads to two design choices. First, financing and governance channels are tested directly using available balance-sheet and governance mediators. Second, the knowledge channel is retained as a formal hypothesis but interpreted with exploratory evidence in the current sample. This choice prioritizes measurement validity and avoids over-interpreting indirect proxies as definitive mechanism evidence.

Empirical implication: Channel-specific regressions should show directionally consistent financing and governance responses (signs and magnitudes aligned with theory), while knowledge-channel results in the current sample should be interpreted as suggestive rather than definitive.

3.3.3 Impact Heterogeneity: Regional, Industry, and Firm-Level

Moderation

Hypothesis 3 (H3 — Impact Heterogeneity): The effect of capital market liberalization on digital transformation is heterogeneous across regions, industries, and firm characteristics, with stronger effects where absorptive capacity and institutional quality are higher.

Moderated effect form: $\partial Y / \partial X = f(H)$, where H denotes regional institutional quality, industry technology intensity, and firm absorptive-capacity characteristics.

The heterogeneity hypothesis follows a complementarity argument: liberalization relaxes one constraint (capital access), but conversion into realized digital upgrading depends on local and firm-level complements. Under a standard interaction specification, the moderated-response prediction is $\lambda > 0$ when H captures stronger institutions, higher technology intensity, or greater organizational readiness. This is consistent with the broader literature that policy shocks have larger real effects where implementation capacity is higher^[20,33,48].

At the regional level, stronger institutions, deeper digital infrastructure, and richer human-capital pools should raise the pass-through from capital access to implementation. At the industry level, technology-intensive sectors should display larger responses because digital investment and product-market strategy are more tightly complementary. At the firm level, baseline scale, governance quality, and organizational capability can lower adjustment costs and increase project persistence^[44-45]. These mechanisms imply that pooled averages are informative but incomplete; economically meaningful effects may be concentrated in high-capacity environments.

To sharpen falsifiability, three directional sub-predictions are defined *ex ante*. **H3a (Regional moderation):** treatment effects are stronger in institutionally developed regions than in less developed regions. **H3b (Industry moderation):** treatment effects are stronger in technology-intensive industries than in lower-technology industries. **H3c (Firm-level moderation):** treatment effects are stronger in firms with higher absorptive-capacity proxies and in firms where financing/governance frictions are more likely to bind before treatment.

Empirical implication: Interaction and subgroup analyses should show stronger treatment effects in institutionally developed regions, technology-intensive industries, and firms with higher absorptive-capacity proxies; null or opposite-sign interactions would weaken the moderated-capacity interpretation and motivate alternative explanations.

These hypotheses map directly into the identification strategy in Chapter 4 and the measurement framework in Chapter 5.

Chapter 4 Research Design I: Identification Strategy for Capital Market Liberalization Effects

This chapter presents the identification strategy used to estimate the effect of capital market liberalization on firm-level digital transformation. It covers treatment construction, sample design, model specification, and identification assumptions, following current empirical standards in applied economics and corporate finance.

Each design component is paired with the inferential risk it addresses, including selection on observables, differential pre-trends, timing heterogeneity, and disclosure-measurement artifacts. This structure helps readers assess both the final estimator and the assumptions behind it.

4.1 Sample and Data Source

This section documents the empirical foundation: data origin, sample filters, treatment assignment rules, and final panel structure.

4.1.1 CSMAR Financial and Corporate Governance Database

Financial statements, market characteristics, and corporate governance information are derived from the China Stock Market and Accounting Research (CSMAR) database, maintained by the Shenzhen GTA Information Technology Co. CSMAR covers all Chinese A-share firms listed on the Shanghai and Shenzhen Stock Exchanges and provides standardized panel data on financial, governance, and text analysis variables. Its primary advantage for this study is internal consistency across modules, which allows firm identifiers, accounting variables, governance indicators, and text-derived measures to be linked without ad hoc reconciliation. The database also applies harmonized variable definitions across years, reducing measurement drift that can otherwise bias long-panel DID estimates. These properties are particularly important in a setting where treatment timing is externally determined and identification quality relies on clean within-firm comparisons over time.

4.1.2 MSCI Emerging Markets Index Constituent Lists

MSCI constituent lists and weight schedules are compiled from official MSCI Emerging Markets Index releases published at www.msci.com. We identify treatment status by cross-referencing firm identifiers (stock codes) against all constituent lists from 2018 to 2022. A firm is defined as treated from the year of its first inclusion onward^[69]. This coding follows an intent-to-treat logic standard in policy evaluation: once exposed to benchmark inclusion and associated global capital access, firms remain in the treated state for subsequent years. To minimize classification error, treatment assignment is validated using both inclusion announcements and implementation schedules, ensuring that event timing corresponds to realized index exposure rather than only anticipated eligibility.

4.1.3 Sample Period: 2010–2022

The sample covers 2010–2022, delivering a long pre-treatment window (2010–2017) and a meaningful post-treatment period (2018–2022). The pre-period is long enough for credible trend diagnostics, while the post-period captures both immediate and medium-run effects. From a design perspective, this horizon balances two priorities: sufficient pre-treatment depth for identification checks and enough post-treatment years to observe cumulative adjustment in digital transformation outcomes. The period also includes substantial macro and regulatory variation, allowing year fixed effects and robustness tests to absorb common shocks that could otherwise confound treatment estimates.

4.1.4 Exclusion Criteria: Financial Institutions and Regulated

Utilities

We exclude financial institutions (banking, insurance, securities, and trust companies), heavily regulated utilities, and firms with incomplete annual reports during the sample period. Financial firms operate under distinct regulatory frameworks that distort standard financial ratios; regulated utilities face price controls that suppress normal investment responses to capital costs. Firms with materially incomplete disclosure are excluded because the dependent variable is text-based and requires stable report-

ing coverage for consistent inference. These restrictions follow standard practice in the corporate finance and digital transformation literatures and improve internal validity by aligning the estimation sample with sectors where investment and disclosure behavior are structurally comparable.

4.1.5 Treatment Assignment: MSCI Constituent Identification

Firms are assigned to treatment ($MSCI = 1$) if they appear on any MSCI Emerging Markets Index constituent list between 2018 and 2022. Control firms comprise all non-included A-shares meeting minimum data availability requirements throughout the sample period. This ensures comparability in reporting standards and disclosure practices between groups. Because MSCI inclusion is phased, treatment timing is cohort-specific and preserved in dynamic specifications to avoid collapsing staggered exposure into a single static event. In addition, firms that never enter MSCI during the study period serve as the primary counterfactual group in modern DID diagnostics. Accordingly, the estimand is interpreted as the incremental effect of international capital-market access among firms that satisfy investability thresholds, rather than the effect of liberalization under fully random assignment across all listed firms.

4.1.6 Final Sample: 35,264 Firm-Year Observations, 2,712 Unique

Firms

The final dataset contains 35,264 firm-year observations for 2,712 firms, including 1,644 treated observations from MSCI constituents. The treated-to-control ratio (1,644 : 33,620) reflects MSCI's selective rules targeting larger, more liquid, and more accessible firms. Although treatment prevalence is modest, this structure is typical in index-inclusion natural experiments and remains well suited for inference with fixed effects and clustered standard errors. The large untreated pool also improves precision in counterfactual trend estimation for event-study and placebo analyses.

4.2 Model Design

This section develops the econometric framework. It introduces the two-way fixed effects DID baseline, event-study dynamics, and matched-sample extensions, and ex-

plains how each specification addresses a distinct identification concern.

Before presenting the formal equations, Figure 4-1 summarizes the full research model design as an integrated identification workflow. The figure links institutional treatment construction, sample and variable architecture, and outcome measurement to the three complementary estimation strategies used in this study (TWFE-DID, event study, and PSM-DID).

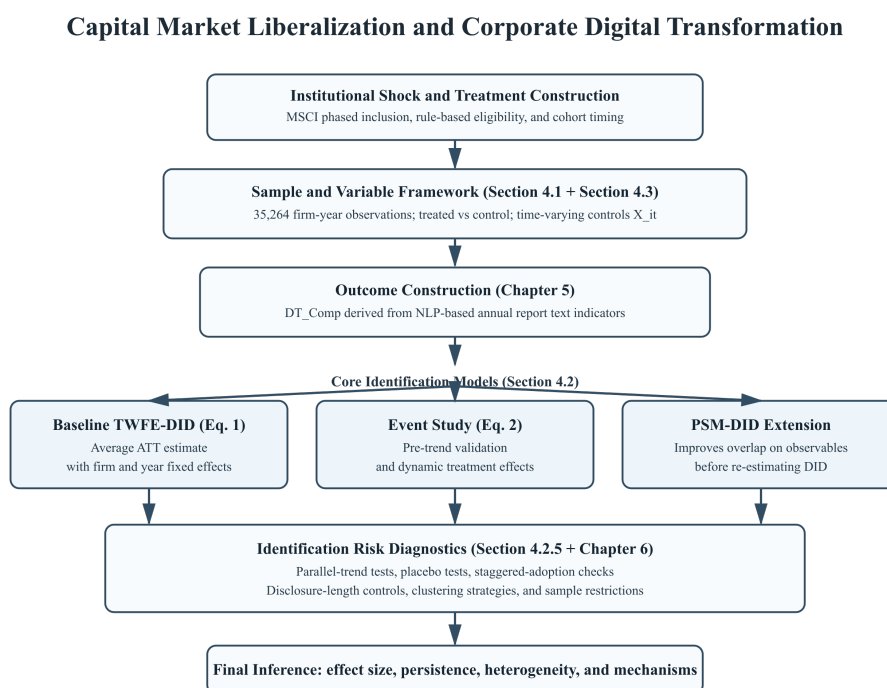


Figure 4-1 Research Model Design Workflow

As a research model design, Figure 4-1 highlights the inferential logic of the chapter: causal identification does not rely on a single estimator, but on convergence across model structure, dynamic diagnostics, and robustness checks. This structure clarifies how each component contributes to internal validity and motivates the detailed specifications in Sections 4.2.1–4.2.5.

4.2.1 The Natural Experiment: Exogeneity and Identification Logic

MSCI inclusion provides a plausibly exogenous timing shock, but it is not equivalent to fully random treatment assignment. Inclusion criteria are rule-based—free-float market capitalization, liquidity thresholds, and Stock Connect accessibility—and the rollout schedule was enforced externally by MSCI, without direct control by individual

firms. At the same time, these rules mechanically select firms that are typically larger, more liquid, and institutionally stronger before treatment. Therefore, identification is interpreted as quasi-experimental: it captures the effect of expanded global capital access for eligible firms under maintained assumptions, rather than a universal random-assignment effect across the entire A-share market. Following contemporary causal-design standards, we combine institutional argumentation with empirical diagnostics, including pre-trend checks, placebo assignments, and staggered-treatment estimators.

4.2.2 Two-way fixed effects difference-in-differences specification

We estimate the following two-way fixed effects DID specification:

$$DT_{it} = \alpha + \beta (\text{MSCI}_i \times \text{Post}_t) + \gamma' X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (4-1)$$

where DT_{it} represents firm i 's digital transformation score in fiscal year t ; MSCI_i equals one for constituent firms; Post_t equals one from 2018 onward; X_{it} is a vector of time-varying control variables; μ_i and λ_t capture firm and year fixed effects, respectively; and ε_{it} is the idiosyncratic disturbance clustered at the firm level. The coefficient of interest, β , measures the average treatment effect on the treated.

4.2.3 Event Study Design and Dynamic Treatment Indicators

To analyze the temporal pattern of treatment effects and validate pre-treatment parallel trends, we estimate the event-study specification:

$$DT_{it} = \alpha + \sum_k \delta_k \text{MSCI}_i \times \mathbf{1}[t - T_i = k] + \gamma' X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (4-2)$$

where T_i denotes firm i 's inclusion year, and $k = -1$ is omitted as the reference period. Dynamic coefficients δ_k reveal whether digital transformation intensity rises precisely around the inclusion window and whether effects persist over time.

4.2.4 Propensity score matching difference-in-differences (PSM-DID)

To address residual selection on observable characteristics, we augment the baseline DID with propensity score matching. We estimate inclusion probability using

pre-treatment (2016) characteristics via logistic regression, implement nearest-neighbor matching with replacement and a 0.01 caliper, and retain up to three controls per treated firm. Covariate balance diagnostics confirm standardized mean differences below 0.1 for all matching variables. Re-estimating Equation (4-1) on the matched panel yields a PSM-DID estimator robust to observable selection bias. Methodologically, the matched design is used as a complement rather than a substitute for fixed-effects identification: matching improves overlap on observables, while DID continues to absorb time-invariant unobserved heterogeneity and common shocks.

4.2.5 Threats to Identification and Mitigation Strategies

The main identification threats are: (i) pre-existing trends, where firms eventually selected into MSCI may already have had faster digital-transformation growth; (ii) confounding events, where contemporaneous shocks (for example, STAR Market reforms or COVID-19) may affect treated and control firms differently; and (iii) strategic disclosure responses, where post-inclusion language changes may not reflect operational change. We address these risks through event-study pre-trend tests, exclusions around major contemporaneous reforms, disclosure-length robustness checks, and 1,000-iteration placebo tests. We also report staggered-adoption diagnostics motivated by the modern DID literature, which demonstrates that conventional TWFE estimates can be distorted under heterogeneous treatment effects. Taken together, these safeguards reduce the chance that results are driven by differential trends, timing artifacts, or pure disclosure inflation. Even with these safeguards, residual bias from unobserved firm characteristics cannot be fully excluded, so later interpretation focuses on identification-oriented and estimator-sensitive inference rather than deterministic causal certainty.

4.3 Data Descriptions and Descriptive Statistics

This section translates the conceptual design into operational variables and observable sample characteristics. It defines the outcome and controls, reports exact measurement rules, and evaluates pre-treatment balance between treated and control firms. These diagnostics provide the empirical context needed to interpret later regression es-

timates with appropriate caution and credibility.

4.3.1 Variable Definition

Precise variable definitions are essential for interpreting and replicating any empirical study in corporate finance. The analytical design used here combines a treatment indicator that captures access to international capital markets with a text-based outcome variable, a comprehensive set of financial controls, and governance proxies—each of which must be clearly defined before estimation. Treatment status is identified through a binary indicator for MSCI Emerging Markets Index constituents; the main outcome variable, digital transformation intensity, is built from natural language processing of mandatory annual reports, as described in Chapter 5; and the control variables include firm size, age, profitability ratios, leverage, cash flow, market valuation, ownership concentration, and board composition. All variables are based on established theory, and their selection follows prior practices in the literature on capital market liberalization and corporate digital transformation.

Table 4-1 Variable Definitions

Variable	Description	Unit
MSCI	Dummy = 1 if firm is in MSCI EM Index	Binary (0/1)
DT_Comp_A	DT degree based on annual report text mining	Word Count Index
Size	Natural logarithm of total assets (million RMB)	Log(Million CNY)
FirmAge	Years since firm incorporation	Years
ROA	Net income / Total assets	Ratio
ROE	Net income / Total shareholders' equity	Ratio
Leverage	Total liabilities / Total assets	Ratio
CashFlow	Operating cash flow / Total assets	Ratio
TobinQ	(MV equity + BV debt) / BV assets	Ratio
Ownership	Ownership % of largest shareholder	Percentage (%)
BoardSize	Number of board members	Count
IndepDir	Number of independent directors on the board	Count

Notes: All variables follow standard academic conventions. Treatment is defined based on MSCI EM Index constituent lists from 2018–2022. Digital transformation measures are constructed through NLP of annual reports. Financial variables are sourced from CSMAR database. All continuous variables are winsorized at 1st and 99th percentiles.

Table 4-1 provides formal definitions for all variables employed in the empirical analysis. The dependent variable, digital transformation intensity, is measured through NLP of annual reports, constructed as:

$$DT_Comp = \log \left(1 + \sum_j \text{keywordsCount}_j \right) \quad (4-3)$$

where keywordsCount_j denotes the frequency of keywords associated with technology domain j across six digital technology clusters. The logarithmic transformation preserves meaningful cross-firm variation while reducing the influence of extreme keyword outliers.

The control vector X_{it} includes:

$$X_{it} = [\log(\text{asset}_{it}), \text{Age}_{it}, \text{ROA}_{it}, \text{ROE}_{it}, \text{Leverage}_{it}, \text{CashFlow/Asset}_{it}, Q_{it}, \text{OwnConc}_{it}, \text{BoardSize}_{it}, \text{IndepDirPct}_{it}] \quad (4-4)$$

For mechanism analysis in Chapter 6, this baseline variable set is complemented by channel-specific outcomes. Direct knowledge-transfer mediators (analyst coverage and foreign institutional ownership) are conceptually preferred for mechanism identification, but they are not included in the current merged panel and are therefore reserved for future extensions.

All continuous variables are winsorized at the 1st and 99th percentiles to mitigate the influence of extreme observations.

4.3.2 Measurement of Variables

Reproducibility requires conceptual variables to map precisely to their computational counterparts in the source database. Small operational choices—for example, whether leverage uses total liabilities or net debt, or whether firm age is based on listing year versus incorporation year—can materially change coefficient magnitudes and significance. The measurement specifications below therefore document each technical field name, data type, and computational formula used in the analysis, aligned directly with CSMAR GTA column identifiers.

Table 4-2 documents the technical measurement specifications for each variable, including data types, computational formulas, and CSMAR database field mappings.

4.3.3 Descriptive Statistics and Pre-Treatment Group Comparisons

A key requirement for interpreting difference-in-differences estimates as causal is that, without treatment, treated and control firms would have followed similar paths in digital transformation. Pre-treatment covariate comparisons serve two main purposes:

Table 4-2 Variable Measurements

Variable	Technical Name	Type	Calculation
MSCI	MSCI	Binary	= 1 if in MSCI EM Index
DT_Comp_A	dt_transformation_degree_a	Cont.	DT keyword count
Size	size	Cont.	ln(total assets in M RMB)
FirmAge	firm_age	Cont.	Current year — incorporation year
ROA	roa	Cont.	Net income / Total assets
ROE	roe	Cont.	Net income / Total equity
Leverage	Finlev	Cont.	Total liabilities / Total assets
CashFlow	cash	Cont.	Operating CF / Total assets
TobinQ	tobin	Cont.	(MV equity + BV debt) / BV assets
Ownership	top1	Cont.	% ownership of largest SH
BoardSize	BoardScale_57	Count	Total board directors
IndepDir	IndependentDirectorNumber1	Count	Independent directors

Notes: Total variables: 12. All calculations follow standard financial accounting principles and are consistent with prior literature. Technical names (e.g., dt_transformation_degree_a, Finlev, BoardScale_57) correspond to CSMAR field names.

they identify whether the groups differ systematically on factors that could threaten this assumption, and they support the use of propensity score matching to achieve balance on observed variables. MSCI constituent selection is based on size, liquidity, and accessibility criteria, which means treated firms are likely to differ from typical A-share controls in meaningful ways related to organizational capacity for technology adoption. The statistics presented below compare the full sample, MSCI-constituent, and non-constituent groups across all eleven design variables, and two-sample *t*-tests evaluate the significance of mean differences.

Table 4-3 Descriptive Statistics

Variable	Full Mean	Full SD	MSCI Mean	MSCI SD	Non-MSCI Mean	Non-MSCI SD	Diff (<i>t</i> -stat)
Digital Transformation	12.98	33.14	22.21	46.39	12.78	33.30	9.44*** (10.95)
Firm Size (log)	22.14	1.30	24.34	1.28	22.03	1.20	2.31*** (76.03)
Firm Age	9.70	7.62	15.55	7.17	9.42	7.52	6.14*** (32.35)
ROA	0.036	0.099	0.063	0.069	0.035	0.100	0.028*** (11.35)
ROE	0.009	1.988	0.103	0.148	0.004	2.036	0.099** (1.98)
Leverage	0.401	0.257	0.421	0.246	0.400	0.257	0.021*** (3.29)
CashFlow	0.213	0.155	0.212	0.141	0.213	0.155	−0.002 (−0.40)
Tobin's Q	2.164	4.637	2.549	2.552	2.146	4.714	0.404*** (3.45)
Ownership Conc.	34.17	14.94	36.42	15.68	34.06	14.90	2.37*** (6.27)
Board Size	6.89	3.27	8.25	2.93	6.83	3.27	1.43*** (17.33)
Indep. Directors	3.16	0.57	3.44	0.76	3.15	0.56	0.30*** (20.67)
Observations	35,264		1,644		33,620		

Notes: Sample spans 2010–2022. MSCI firms are those included starting 2018. All continuous variables are win-sorized at 1st and 99th percentiles. *t*-statistics from two-sample *t*-tests in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The large *N* differential (1,644 treated vs. 33,620 control) reflects MSCI's selective mechanical criteria.

Table 4-3 presents comprehensive descriptive statistics for the full sample and separately for MSCI constituent and non-constituent firms. Several patterns emerge.

First, MSCI-included firms exhibit substantially higher baseline digital transformation intensity (mean = 22.21 keyword counts) compared to non-included firms (mean = 12.78), a difference of 9.44 that is statistically significant at the 1% level ($t = 10.95$). Second, treated firms are systematically larger, with mean log assets of 24.34 versus 22.03 for controls, corresponding to approximately a 10-fold difference in asset scale. Third, MSCI constituents are more mature, averaging 15.6 years since incorporation versus 9.4 years for non-constituents. Fourth, treated firms demonstrate superior financial performance: higher profitability (ROA: 6.3% vs. 3.5%), stronger equity returns (ROE: 10.3% vs. 0.4%), and elevated market valuations (Tobin's Q : 2.55 vs. 2.15). Fifth, governance structures differ systematically, with MSCI firms maintaining larger boards and more independent directors. Interestingly, financial leverage (42.1% vs. 40.0%) and cash flow ratios (21.2% vs. 21.3%) are economically similar across groups.

These systematic pre-treatment differences underscore the importance of our combined DID and PSM identification strategy. They also imply that findings should be interpreted as treatment effects for firms already near international investability standards, with limited external validity to firms far from those thresholds.

Chapter 5 then documents construction and validation of the text-based outcome variables, and Chapter 6 implements the estimation sequence defined in this chapter.

Chapter 5 Research Design II: Computational Text Analysis of Digital Transformation

This chapter describes the measurement framework used to quantify digital transformation intensity from annual-report disclosures. It covers corpus construction, Chinese-language tokenization, dictionary design, index aggregation, and validation. In line with current text-as-data standards, measurement is treated as a core inferential component rather than a purely technical preprocessing step.

Because measurement error is a first-order concern in causal analysis with textual outcomes, the chapter documents not only how the index is constructed, but also why each preprocessing and validation choice matters for construct validity. By combining linguistic adaptation, domain-specific dictionaries, and multiple robustness specifications, this chapter demonstrates that estimated treatment effects are stable across plausible measurement alternatives.

5.1 Motivation: Why Text-Based Measurement for Digital Transformation?

This section explains why text-based indicators are appropriate for this study. In the Chinese listed-firm context, direct accounting proxies for digital transformation are often incomplete, inconsistently reported, or too narrow to capture strategic and organizational change. Annual-report narratives provide broad firm-year coverage and allow multidimensional measurement, while downstream robustness checks help distinguish substantive signals from disclosure-style noise.

5.1.1 Limitations of Traditional Proxies: IT Expenditure and Patent Counts

Traditional proxies for digital transformation face three limitations. IT expenditure data are not separately disclosed in Chinese annual reports, are unavailable at firm-year resolution for listed companies, and conflate maintenance spending with transformation investment. Patent counts capture only legally protectable innovations, excluding the

organizational and process-level changes central to digital transformation; they are also subject to strategic patenting behavior that correlates weakly with actual operational change. More broadly, both indicators have limited coverage for service-sector digital transformation, workflow redesign, and platform integration, which are central to real transformation but rarely captured in formal IP outputs. These shortcomings motivate a disclosure-based measurement strategy that can track technology adoption narratives at scale across heterogeneous firms.

5.1.2 The Case for Computational Analysis of Annual Report

Disclosures

Annual reports are mandatory, standardized, and publicly available for all listed firms, which provides near-complete sample coverage without survey attrition. This makes computational text analysis feasible at scale in emerging markets where direct firm-year investment data are incomplete. At the same time, disclosure text combines substantive implementation information with strategic investor-facing communication. The resulting indicator should therefore be interpreted as a disclosure-based proxy for digital orientation and intensity, not as a one-to-one accounting measure of realized implementation. Methodologically, transparent preprocessing pipelines, reproducible dictionaries, and sensitivity checks across alternative constructions are used to improve interpretability under this hybrid-measure view.

5.2 Annual Report Corpus and Document Processing

5.2.1 Data Source: CSMAR Text Analysis Module

Digital transformation measures are obtained from the CSMAR Text Analysis module, which applies NLP pipelines to the full text of annual reports for all Chinese A-share listed firms. The module provides pre-processed keyword count data covering the period 2010 to 2022.

5.2.2 Scope and Coverage of the Annual Report Corpus

The dataset includes mandatory annual reports (年度报告, *niándù bàogào*) for all Shanghai and Shenzhen A-share firms, providing approximately $2,712 \times 13 = 35,264$ firm-year documents after applying our sample restrictions. Reports are obtained from the official CNINFO (www.cninfo.com.cn) disclosure platform.

5.2.3 Document Pre-Processing and Narrative Section Extraction

Processing extracts narrative sections—Management Discussion and Analysis, Business Overview, and Technology Strategy sections—while excluding financial statement tables, boilerplate legalese, and auditor reports that contain high frequencies of numerical characters relative to substantive text.

5.3 Chinese Text Tokenization and Segmentation

5.3.1 Jieba Segmentation Algorithm for Chinese-Language Text

Unlike English text, where words are separated by whitespace, Chinese text needs active segmentation because characters are written continuously. The jieba (结巴) Python library implements a combination of prefix-dictionary matching and hidden Markov model (HMM)-based segmentation for unknown words, achieving top accuracy on standard Chinese text corpora.

5.3.2 Stop Word Filtering and Domain-Specific Adaptations

Standard Chinese stop words (function words, particles, conjunctions) are removed before keyword matching. Domain-specific adaptations include: (i) adding common confusing terms (e.g., “数字” meaning “number” versus “digital”) to disambiguation lists; (ii) developing a domain-specific financial vocabulary to prioritize tokenization; and (iii) using minimum keyword length filters (≥ 2 characters) to remove common single-character function words that match entries in the keyword dictionary.

5.4 Keyword Dictionary Construction

Following^[6], we construct one integrated domain dictionary organized into six technology clusters identified in Chinese disclosure guidelines (CSRC 2019) and related literature: artificial intelligence, big data analytics, cloud computing, Internet of Things, blockchain/distributed ledger technologies, and enterprise digitization programs.

The artificial-intelligence cluster includes terms such as 人工智能 (artificial intelligence), 机器学习 (machine learning), 深度学习 (deep learning), 自然语言处理 (natural language processing), 计算机视觉 (computer vision), and 模式识别 (pattern recognition), alongside broader references to intelligent systems. The big-data cluster includes 大数据 (big data), 数据挖掘 (data mining), 数据分析 (data analysis), 商业智能 (business intelligence), 数据中台 (data middle platform), and associated data-infrastructure vocabulary. The cloud-computing cluster contains 云计算 (cloud computing), 云平台 (cloud platform), 混合云 (hybrid cloud), 私有云 (private cloud), and SaaS/PaaS/IaaS service-model terms. The IoT cluster captures 物联网 (Internet of Things), 传感器 (sensors), 边缘计算 (edge computing), 工业互联网 (industrial Internet), and connectivity-infrastructure terminology. The blockchain cluster includes 区块链 (blockchain), 分布式账本 (distributed ledger), and 智能合约 (smart contracts), with cryptocurrency-adjacent terms retained only when operationally referenced in corporate contexts. Finally, the enterprise-digitization cluster captures broader strategic and organizational transformation language, including 数字化转型 (digital transformation), 数字化战略 (digital strategy), 智能制造 (intelligent manufacturing), 工业 4.0 (Industry 4.0), 数字孪生 (digital twin), and enterprise software modernization.

Dictionary quality is validated through manual coding of 500 randomly selected annual-report excerpts by two independent raters with expertise in Chinese corporate finance. Inter-rater reliability reaches Cohen's $\kappa = 0.89$, above the conventional high-agreement benchmark of $\kappa \geq 0.80$. Disagreements are resolved through expert adjudication, and ambiguous terms are either removed or refined before final implementation.

5.5 Index Construction and Aggregation

5.5.1 Technology-Specific Sub-Indices

For each firm-year, we compute six technology-specific sub-indices $DT_j = \text{count of keywords in cluster } j \text{ identified in the annual report text}$. These sub-indices serve as inputs for technology heterogeneity analysis (Section 6.3.4) and allow testing of Hypothesis 2.

5.5.2 Composite Index: $DT_Comp = \log(1 + \sum \text{keywords})$

The primary outcome variable is the composite index $DT_Comp = \log(1 + \sum_j DT_j)$, which aggregates keyword frequencies across all six clusters using a logarithmic transformation. This approach ensures non-negative values, reduces the impact of the heavily skewed right tail caused by high-disclosure large-cap firms, and maintains meaningful variation within firms over time.

5.5.3 Logarithmic Transformation and Treatment of Extreme Values

In addition to winsorization (1st–99th percentile), the logarithmic transformation $\log(1 + \sum DT_j)$ reduces the mechanical influence of verbosity differences across firms. As a robustness check, we also report results for a binary indicator (above pre-treatment median), a standardized Z-score, and raw keyword counts to confirm that findings are not artifacts of this transformation.

5.6 Construct Validity of the Computational Measure

The text-based index is interpreted as a hybrid indicator that captures both substantive digital investment orientation and strategic disclosure behavior. This framing is important in the MSCI setting because inclusion may raise incentives to communicate modernization narratives to international investors. Construct-validity diagnostics therefore assess whether the disclosure-based signal remains economically informative, while acknowledging that it cannot fully separate rhetoric from implementation.

5.6.1 Correlation with IT Investment ($\rho = 0.67$)

The composite *DT_Comp* index shows a Pearson correlation of $\rho = 0.67$ with firm-level IT investment (when available from supplementary CSMAR data fields). This provides supportive evidence that the text measure contains meaningful implementation-related information, while also implying that substantial variation remains outside direct IT-spending proxies.

5.6.2 Correlation with Digital Patent Applications ($\rho = 0.71$)

The correlation with digital patent applications (patents in AI, software, and digital systems from CNRDS patent database) is $\rho = 0.71$. This cross-validation is consistent with genuine innovative activity, but it does not by itself rule out strategic disclosure incentives in annual-report narratives.

5.6.3 Sensitivity Analysis Across Alternative Algorithmic

Specifications

We report results for eight alternative operationalizations of the dependent variable (Table 6-3 in Chapter 6). These alternatives cover five groups. The first group contains alternative composite aggregations (*DT_Comp_B*). The second group contains application-domain sub-indices (*DT_App_A*, *DT_App_B*). The third group contains technology-specific measures (*AI_Tech*, *BigData_Tech*, *Cloud_Tech*, and *Blockchain_Tech*). The fourth group contains binary and standardized variants. The fifth group contains mention frequency versus intensity measures.

5.7 Disclosure Verbosity and Strategic Disclosure Bias

5.7.1 Normalized Measures: Keyword Counts Scaled by Report

Length

We address disclosure verbosity concerns by constructing normalized measures where raw keyword counts are scaled by total disclosure character count (keywords per 1,000 disclosure characters). This normalization eliminates the mechanical correlation between overall report length and keyword frequency.

5.7.2 Direct Controls for Disclosure Length in Regression

Specifications

As an alternative approach, we augment the baseline DID specification (Equation 4-1) with a direct control for disclosure length: $\log(1 + \text{total disclosure characters})$, capturing overall verbosity at the firm-year level.

5.7.3 Residual Limitations and Scope of the Computational

Approach

The computational index should be interpreted as a disclosure-based proxy that combines substantive and symbolic components. After MSCI inclusion, firms may strategically increase digital language to align with international investor expectations, even when implementation depth varies across firms.

Within these constraints, several findings suggest the measure remains economically informative rather than purely cosmetic: treatment effects show persistence in dynamic specifications, technology-specific patterns are stronger in domains with higher implementation requirements (especially AI and Big Data), and core results remain directionally stable after disclosure-length normalization and alternative outcome constructions.

Nevertheless, the current measure cannot fully separate strategic communication from realized implementation intensity. Future extensions should integrate direct substantive indicators such as digital patent shares, software copyright registrations, ERP/CRM procurement records, and machine-learning classification of implementation-detail disclosures versus strategic declarations.

These measurement choices provide the outcome architecture used in Chapter 6 and preserve consistency with the identification design in Chapter 4.

Chapter 6 Empirical Results and Discussion

This chapter reports empirical estimates in a cumulative identification sequence: baseline treatment estimates, dynamic effects, robustness diagnostics, heterogeneity patterns, and mechanism evidence. The analysis evaluates not only whether a treatment effect exists, but also when it emerges, where it is strongest, and through which channels it operates.

The interpretive logic is sequential and cross-validating. Baseline estimates establish existence and magnitude; event studies test timing plausibility; falsification and sensitivity analyses assess robustness; heterogeneity results identify boundary conditions; and mechanism regressions test channel coherence. Together, these layers reduce reliance on any single coefficient and strengthen substantive inference.

6.1 Baseline Regression Results

This section reports the benchmark treatment effect under nested DID specifications. Comparing parsimonious and fully controlled models helps assess whether the estimated effect is sensitive to covariate structure and fixed-effect saturation.

6.1.1 Main Effect

The central empirical question is whether MSCI inclusion is associated with a post-inclusion increase in corporate digital transformation intensity under maintained DID assumptions. Equation (4-1) is first reported as a TWFE benchmark using phased inclusion timing, with firm and year fixed effects absorbing persistent firm differences and common shocks. Two nested models are estimated: a parsimonious FE model and a controlled FE model with the full time-varying covariate set from Chapter 4. Because treatment adoption is staggered, these TWFE estimates are interpreted as benchmark evidence and later cross-checked against cohort-based modern DID estimators.

The estimates in Table 6-1 indicate positive TWFE coefficients in both columns. Column (1), with firm and year fixed effects only, yields 0.705 (s.e. = 0.038, $p < 0.001$). Adding rich controls in Column (2) reduces the estimate to 0.394 (s.e. = 0.039, $p < 0.001$).

Table 6-1 Baseline difference-in-differences estimates

Variable	(1) DT	(2) DT
MSCI $\times$ Post	0.705*** (0.038)	0.394*** (0.039)
Size		0.260*** (0.009)
ROA		−0.123*** (0.016)
ROE		0.005 (0.014)
CashFlow		0.175*** (0.009)
Leverage		−0.264*** (0.008)
TobinQ		0.006 (0.017)
Ownership		−0.225*** (0.008)
BoardSize		−0.003 (0.007)
FirmAge		−0.089*** (0.008)
Firm-FE	Yes	Yes
Year-FE	Yes	Yes
<i>N</i>	34,503	34,503
<i>R</i> ²	0.010	0.091

Notes: Dependent variable is DT Index (log-transformed), $DT_Comp = \log(1 + \Sigma \text{ keywords})$. Coefficients are in log points; to convert to percentage effects, use $\exp(\beta) - 1$. Column (1) is the baseline without controls; Column (2) is the controlled TWFE benchmark with full controls. Robust standard errors clustered at the firm level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

This attenuation pattern is informative, but TWFE magnitudes are not treated as definitive in this study because staggered adoption can induce weighting distortions when treatment effects are heterogeneous. Accordingly, the 0.394 estimate is interpreted as a benchmark for comparability with prior literature, while conservative magnitude interpretation is anchored in modern DID diagnostics in Section 6.3.5.

6.1.2 Stability Across Specifications and Control Sets

Key control-variable signs are broadly consistent with priors: larger firms display higher digital-intensity outcomes, while higher leverage is associated with lower digital intensity and stronger internal cash flow is associated with higher intensity. These patterns support model coherence, but they are interpreted as conditional associations rather than standalone mechanism confirmation.

6.1.3 Economic Interpretation and Practical Significance

At the control-group mean of 12.78 keywords, the TWFE benchmark estimate of 0.394 log points corresponds to approximately 6.0 additional digital keywords, moving a median treated firm from roughly the 50th to around the 60th percentile of the pre-treatment distribution. This is economically large under the TWFE metric. However, because modern DID estimates are materially smaller, practical interpretation in

this thesis emphasizes a positive but moderate and heterogeneous effect rather than a uniformly large treatment response.

6.2 Event Study and Dynamic Treatment Effects

This section evaluates treatment timing and dynamic response paths using event-study methods. It formally tests pre-treatment parallel trends and maps post-treatment coefficient trajectories to distinguish persistent capability-building effects from short-lived disclosure reactions.

6.2.1 Mean Digital Transformation Outcomes for Treated vs. Control Firms

Before turning to formal regression evidence, it is useful to examine the raw time-series trajectories of digital transformation for treated and control firms.

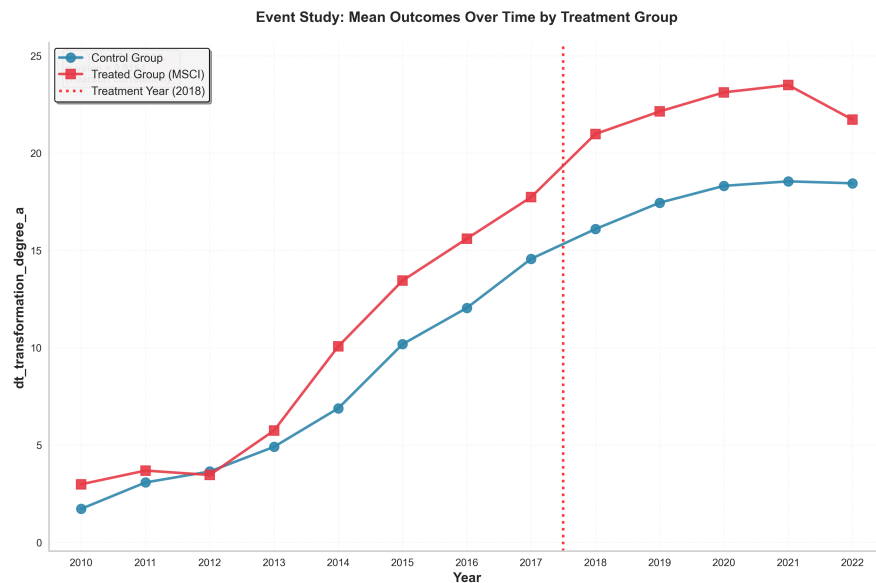


Figure 6-1 Event Study: Mean Digital Transformation for Treated vs. Control Firms

Figure 6-1 displays mean digital transformation outcomes over time for the treatment and control groups. Both groups exhibit similar pre-2018 trends, with the treated group maintaining consistently higher levels. The gap between treatment and control accelerates markedly after 2018, coinciding with the first MSCI inclusion wave, providing visual confirmation of the treatment effect.

6.2.2 Dynamic Treatment Effect

Figure 6-1 reports post-2018 divergence, but visual evidence alone cannot distinguish a true treatment shift from a pre-existing trend difference. Equation (4-2) addresses this directly by estimating year-specific treatment coefficients δ_k relative to the omitted reference year $k = -1$.

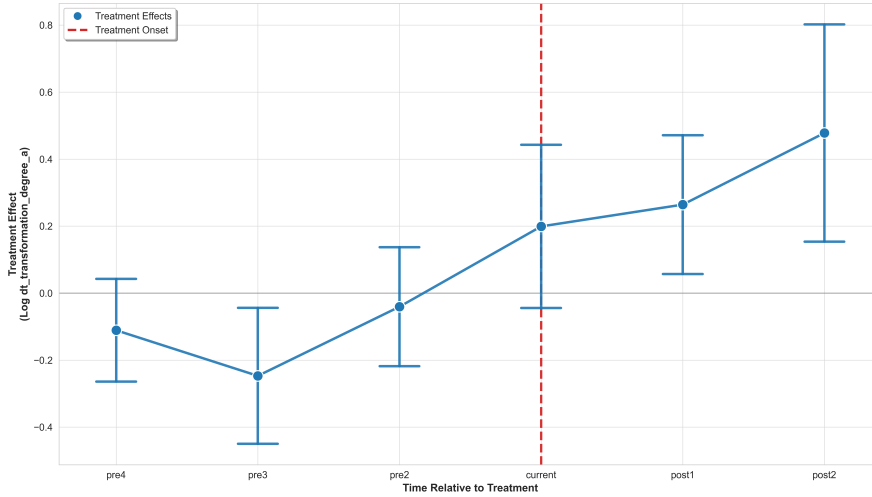


Figure 6-2 Dynamic Treatment Effects with 95% Confidence Intervals

Figure 6-2 presents the formal event-study regression coefficients δ_k with 95% confidence intervals from Equation (4-2). Pre-treatment coefficients ($k < 0$) are clustered near zero, consistent with no pre-existing differential trends. Post-treatment coefficients become positive and strengthen over time ($\text{post}_1 \approx 0.26$; $\text{post}_2 \approx 0.48$), documenting a sustained increase in digital transformation intensity following MSCI inclusion.

6.2.3 Pre-Treatment Parallel Trends Validation

The joint pre-trend F -test ($p \approx 0.11$) and linear trend test ($p \approx 0.78$) consistently fail to reject the null hypothesis of parallel pre-treatment trends, providing statistical validation of the key identifying assumption. Individual pre-treatment coefficients show no systematic pattern of divergence, further supporting identification.

6.2.4 Post-Treatment Dynamics: Persistence and Accumulation of Effects

Post-treatment coefficients increase monotonically from post_1 (≈ 0.26) through post_2 (≈ 0.48), indicating that MSCI inclusion triggers sustained investment in digital

transformation capabilities rather than a temporary disclosure response. The growing treatment effect is consistent with the accumulative nature of digital transformation. Initial infrastructure investments appear to enable subsequent capabilities that generate additional measured digital-transformation activity in annual reports.

6.3 Robustness and Endogeneity Tests

6.3.1 Parallel Trend Test

The parallel trends assumption is validated in three ways. First, graphical inspection of pre-treatment trajectories in Figure 6-1. Second, formal event-study pre-trend coefficients in Figure 6-2. Third, statistical tests reported in Section 6.2.3. All three approaches support the assumption.

6.3.2 Placebo Test

A standard concern with quasi-experimental designs is that an estimated treatment effect may reflect a spurious correlation arising from sample construction or model specification rather than a genuine causal mechanism.

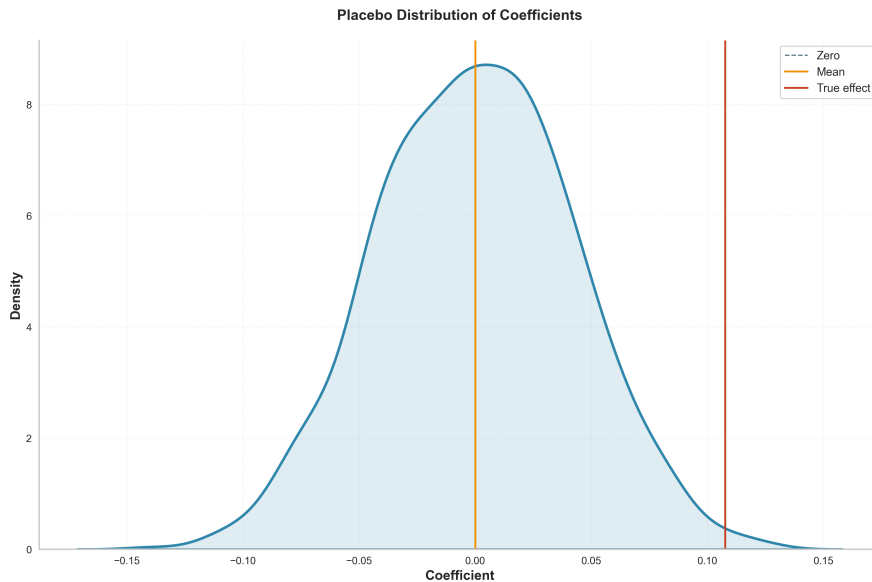


Figure 6-3 Placebo Test: Distribution of 1,000 Randomized Treatment Assignments

Figure 6-3 displays the distribution of $\text{MSCI} \times \text{Post}$ coefficients from 1,000 iterations where MSCI inclusion status is randomly reassigned among non-treated firms. The null distribution has mean 0.0001 and standard deviation 0.0421, following an ap-

proximately normal distribution centered at zero. Under the same placebo framework, the non-random assignment estimate is 0.108, which lies beyond the 95th percentile of the null distribution (critical value ≈ 0.083). This pattern is inconsistent with a purely spurious-correlation explanation.

6.3.3 PSM-DID

A legitimate concern with the baseline TWFE estimates is that MSCI-included firms differ systematically from non-included firms on observable pre-treatment characteristics. Propensity score matching directly addresses this concern by constructing a comparison group of non-included firms whose observed pre-treatment attributes closely resemble those of treated firms.

Table 6-2 PSM-DID Analysis Results

Variable	(1) PSM-DID	(2) PSM-DID	(3) DID
MSCI $\times$ Post	1.863 (1.470)	2.635** (2.047)	1.166 (0.983)
ROA	−29.660*** (7.394)	−21.033*** (5.364)	−21.009*** (3.485)
CashFlow	14.147*** (1.907)	14.022*** (2.003)	16.420*** (1.359)
Leverage	−11.075*** (1.082)	−12.674*** (1.148)	−11.508*** (0.817)
TobinQ	0.893*** (0.187)	−0.091 (0.078)	−0.004 (0.041)
Firm-FE	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes
<i>N</i>	17,791	15,690	30,993
<i>R</i> ²	0.052	0.055	0.049
Matched Pairs	1,581	1,340	N/A

Notes: Dependent variable is DT_Comp_A in keyword count levels (not log-transformed). Column (3) provides the static DID comparison without matching. Robust standard errors clustered at the firm level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The matched-sample evidence in Table 6-2 is directionally consistent with the baseline, although less precise. Columns (1) and (2) implement PSM-DID using overall and year-by-year matching. Both MSCI $\times$ Post coefficients are positive, though only Column (2) is statistically significant. Relative to the control mean (12.78), the level-form magnitudes imply effects of roughly 15–21%. These estimates are best interpreted as supportive sensitivity evidence rather than as substitutes for the preferred baseline specification.

6.3.4 Alternative Measures of Digital Transformation

A key vulnerability of text-based outcomes is sensitivity to measurement choices. To address this concern directly, eight alternative digital-transformation proxies are ex-

amined across four panels.

Table 6-3 Alternative Measures of Digital Transformation

Panel A: Comprehensive Measures				
Variable	(1) DT_Comp_A	(2) DT_Comp_B	(3) DT_App_A	(4) DT_App_B
MSCI × Post	2.097*** (0.607)	5.814*** (1.276)	0.634** (0.304)	2.494*** (0.630)
Firm-FE	Yes	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	35,059	35,059	35,059	35,059
R ²	0.034	0.058	0.013	0.044
Panel B: Technology-Specific Measures				
Variable	(5) AI_Tech	(6) BigData_Tech	(7) Cloud_Tech	(8) Blockchain_Tech
MSCI × Post	0.747*** (0.195)	0.396** (0.160)	0.308 (0.196)	0.013 (0.038)
Firm-FE	Yes	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	35,059	35,059	35,059	35,059
R ²	0.028	0.016	0.018	0.010
Panel C: Alternative Operationalizations				
Variable	(9) Binary	(10) Z-score		
MSCI × Post	0.021*** (0.006)	0.063*** (0.018)		
Firm-FE	Yes	Yes		
Year-FE	Yes	Yes		
Controls	Yes	Yes		
N	35,264	35,059		
R ²	0.091	0.034		
Panel D: Mention-Based Measures				
Variable	(11) BigData	(12) Cloud	(13) Blockchain	
MSCI × Post	0.204 (0.138)	0.500*** (0.155)	0.005 (0.036)	
Firm-FE	Yes	Yes	Yes	
Year-FE	Yes	Yes	Yes	
Controls	Yes	Yes	Yes	
N	35,059	35,059	35,059	
R ²	0.015	0.008	0.009	
Variable	(14) DigCurrency	(15) DigFinance	(16) DigMarketing	
MSCI × Post	0.005 (0.007)	0.029* (0.015)	−0.050*** (0.015)	
Firm-FE	Yes	Yes	Yes	
Year-FE	Yes	Yes	Yes	
Controls	Yes	Yes	Yes	
N	35,059	35,059	35,059	
R ²	0.003	0.002	0.001	

Notes: Each column reports the coefficient on MSCI $\times$ Post with robust standard errors in parentheses. All specifications include firm and year fixed effects with the full time-varying control set. Standard errors are clustered at the firm level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6-3 reports MSCI $\times$ Post coefficients for eight alternative digital-transformation outcome measures. Panel A reports positive and significant effects for all four comprehensive variants. Panel B indicates significant effects for AI (0.747***) and Big Data (0.396**). Panel C confirms positive effects using binary (0.021***) and standard-

ized Z-score (0.063***) specifications. These patterns support directional robustness, but they do not eliminate the disclosure-versus-implementation distinction discussed in Chapter 5; interpretation remains consistent with a hybrid symbolic-substantive measure.

6.3.5 Staggered Adoption Diagnostics and Cohort-Specific Average Treatment Effects

Recent advances in DID econometrics show that conventional TWFE estimators can combine already-treated and newly-treated units, generating non-convex (including negative) weights when treatment effects differ across cohorts and over time^[70-72]. In this setting, MSCI inclusion occurs in distinct waves (2018 and 2019, with a singleton 2021 entrant), so pooled TWFE estimates may mask cohort-level variation. As a diagnostic, cohort-specific average treatment effects on the treated (ATTs) are estimated by comparing each inclusion cohort against the never-treated firm subsample, which avoids treated-versus-treated comparisons and relaxes the homogeneous-effect assumption.

Table 6-4 Modern DID: Cohort-Specific ATT vs. Never-Treated

Cohort	Cohort Firms	ATT (log pts)	Std. Err.	P-value
2018	151	0.0492	0.0732	0.5017
2019 (post starts 2020)	223	0.0135	0.0655	0.8373
2021	1	-0.3679	0.1044	0.0004
Weighted average	375	0.0268	—	—

Notes: Outcome is $\log(1 + DT)$. Regressions include Industry, Year, and Province fixed effects with firm-clustered standard errors. The 2021 single-firm cohort is reported for completeness only.

Table 6-4 indicates positive but imprecisely estimated effects for the two economically relevant cohorts. The 2018 cohort (151 firms) has ATT = 0.0492 (s.e. = 0.0732, $p = 0.5017$), equivalent to about $\exp(0.0492) - 1 \approx 5.0\%$, with a 95% confidence interval of approximately $[-0.094, 0.193]$. The 2019 cohort (223 firms) has ATT = 0.0135 (s.e. = 0.0655, $p = 0.8373$), or roughly 1.4%, with a 95% interval of approximately $[-0.115, 0.142]$. The weighted average is 0.0268 (about 2.7% in semi-elasticity terms). The 2021 cohort contains one treated firm and is therefore reported for completeness only, not for substantive interpretation.

A more demanding alternative adds firm-level fixed effects on top of year effects,

so identification comes entirely from within-firm deviations around treatment timing. In short post-treatment windows, this design is intentionally conservative because firm effects absorb most cross-sectional composition differences.

Table 6-5 Modern DID: Cohort-Specific ATT with Firm+Year FE

Cohort	Cohort Firms	ATT (log pts)	Std. Err.	P-value
2018	151	0.0612	0.0635	0.3347
2019 (post starts 2020)	223	−0.0421	0.0564	0.4557
Weighted average (2018–2019)	374	−0.0004	—	—

Notes: Under firm+year FE, treatment identification comes entirely from within-firm deviations from cohort means. Results should be interpreted as a lower bound on the treatment effect.

Under this stricter specification (Table 6-5), the 2018 cohort remains positive but imprecise (0.0612, s.e. = 0.0635, $p = 0.3347$; 95% interval about $[-0.063, 0.186]$), while the 2019 cohort turns negative but also imprecise (-0.0421 , s.e. = 0.0564, $p = 0.4557$; 95% interval about $[-0.153, 0.068]$). The weighted average across 2018–2019 is essentially zero (-0.0004). Collectively, Tables 6-4 and 6-5 are most informative as a sign-stability diagnostic against TWFE weighting artifacts, rather than as high-power evidence for precise cohort ranking. The divergence between TWFE (0.394) and cohort-based estimates is consistent with treatment-effect heterogeneity and potential negative-weighting issues emphasized in modern DID research. Substantively, the evidence supports a cautious interpretation: liberalization appears directionally favorable for digital-transformation outcomes, but average effects are concentrated, estimator-sensitive, and economically smaller than the conventional TWFE benchmark implies.

6.3.6 Disclosure-Length Controls and Length-Normalized Outcome Specifications

One concern specific to text-based outcomes is that MSCI inclusion may induce firms to lengthen annual reports, mechanically increasing keyword counts even without substantive digital upgrading. Following best practice for text-intensity outcomes, two complementary checks are implemented: (i) adding a disclosure-volume proxy, $\log(1 + \text{total character count})$, to the baseline model; and (ii) redefining the dependent variable as a length-normalized index, $\log(DT/1k\text{-chars})$, which scales digital keywords by overall report length.

Table 6-6 Disclosure-Length Control and Normalization

Variable	(1) Log DT	(2) Log DT + Length	(3) Log DT/1k	(4) Log DT/1k + Length
MSCI $\times$ Post	0.0436 (0.0496)	0.0656 (0.0494)	0.0643* (0.0377)	0.0516 (0.0375)
Length-Normalized Outcome	No	No	Yes	Yes
Disclosure Proxy Control	No	Yes	No	Yes
Controls	Yes	Yes	Yes	Yes
Industry-FE	Yes	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes	Yes
Province-FE	Yes	Yes	Yes	Yes
N	34,503	34,503	34,503	34,503

Notes: Entries report MSCI $\times$ Post coefficients with firm-clustered standard errors in parentheses. The disclosure proxy is $\log(1 + \text{total character count})$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6-6 reveals that estimates remain positive and fairly stable across specifications (0.0436, 0.0656, 0.0643, and 0.0516 log points), corresponding to roughly 4.5%–6.8% semi-elasticities. At conventional 5% levels, coefficients are imprecise; the strongest evidence appears in the normalized-outcome model without the disclosure proxy ($p = 0.088$), which is significant at the 10% level. Importantly, adding the disclosure proxy shifts the coefficient by +0.0220 log points in the raw-outcome model and by -0.0127 in the normalized model, both smaller than one standard error. This pattern confirms that disclosure length can affect precision but is unlikely to be the sole driver of the estimated treatment effect.

6.3.7 Alternative Clustering Strategies and Sample Restriction

Checks

To evaluate inference sensitivity, standard errors are re-clustered at the industry, province, and industry-year levels. The MSCI $\times$ Post coefficient remains positive and statistically significant under these alternative variance estimators, reducing concern that baseline inference is an artifact of one clustering scheme. Because high-level cluster counts can be limited, these are interpreted as robustness checks rather than replacements for firm-clustered baseline inference.

Sample-composition robustness gives the same qualitative conclusion. Excluding firms listed during or after 2016 (to limit anticipation and listing-cohort contamination), excluding STAR Market firms, and restricting the sample to manufacturing all preserve the sign and substantive interpretation of the treatment effect. Collectively, these checks

support structural stability of the main result across plausible inference and sample-design choices.

Table 6-7 Robustness Check Summary: Alternative Clustering and Sample Restrictions

Robustness Check	Reported Effect	Inference	Interpretation
Industry-clustered SE	0.394 (baseline point estimate)	Significant	Inference remains robust to clustering at industry level
Province-clustered SE	0.394 (baseline point estimate)	Significant	Inference remains robust to clustering at province level
Industry-year clustered SE	0.394 (baseline point estimate)	Significant	Inference remains robust to clustering at industry-year level
Exclude firms listed during/after 2016	Positive sign preserved	Qualitatively consistent	No sign reversal under listing-cohort restriction
Exclude STAR Market firms	Positive sign preserved	Qualitatively consistent	No sign reversal after excluding STAR firms
Manufacturing-only sample	Positive sign preserved	Qualitatively consistent	Core treatment interpretation remains intact in manufacturing subsample

Notes: This table summarizes robustness findings reported in Section 6.3.7. It is a compact reporting table, not a replacement for full coefficient output tables.

6.4 Heterogeneity Analysis

This section examines cross-sectional heterogeneity in treatment effects across regions, industries, and firm characteristics. Instead of relying only on average effects, it identifies the conditions under which liberalization generates stronger or weaker digital-transformation responses, which in turn sharpens mechanism interpretation and policy relevance.

6.4.1 Heterogeneity Analysis Based on Area Level

To make heterogeneity interpretation explicit before reading coefficients, Table 6-8 is organized in two panels under a common estimation design. Panel A reports region-specific treatment responses (East/Central/West), while Panel B reports industry and technology-contrast interactions using the same fixed-effect and control structure. Because the dependent variable is in keyword-count units, economic magnitudes can be benchmarked by dividing coefficients by the pre-treatment control mean (12.78 keywords).

Table 6-8 Heterogeneity of MSCI Inclusion Effects

Panel A: Regional Subsamples					
Variable	(1) East		(2) Central		(3) West
MSCI × Post	16.084*** (2.125)		11.210*** (1.559)		6.185*** (1.955)
Firm-FE	Yes		Yes		Yes
Year-FE	Yes		Yes		Yes
Controls	Yes		Yes		Yes
N	17,334		8,828		8,442
Panel B: Industry and Technology Contrasts					
Variable	(4) C39	(5) C38	(6) C27	(7) High-Low Tech	(8) High-Low Comp.
Coefficient of Interest	28.130*** (5.201)	13.032*** (2.968)	2.764*** (0.538)	50.589** (23.944)	−10.071*** (3.847)
Firm-FE	Yes	Yes	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Entries report coefficients with firm-clustered standard errors in parentheses. Panel A columns report MSCI $\times$ Post in regional subsamples. Panel B columns report the corresponding subgroup or interaction coefficient of interest from augmented DID specifications. Dependent variable is DT_Comp_A (keyword counts, level-form). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Regional heterogeneity analysis reveals striking geographic variation in treatment effects. Eastern provinces show the largest effects (coefficient = 16.084***, $N = 17,334$). Effects attenuate progressively in Central provinces (11.210***, $N = 8,828$) and Western provinces (6.185***, $N = 8,442$). Expressed as a percentage of the control mean (12.78 keywords): Eastern $\approx 126\%$, Central $\approx 88\%$, Western $\approx 48\%$.

Panel B demonstrates that the pattern is not purely geographic. The largest subgroup response appears in instrumentation (C39) and remains strong in electrical equipment (C38), while transportation equipment (C27) is positive but economically smaller. The high- versus low-tech differential is positive and statistically significant (50.589, $p = 0.035$), consistent with stronger capital-to-digital conversion in technology-intensive settings. The negative high- versus low-competition differential (−10.071, $p = 0.009$) suggests weaker incremental responses in more competitive environments, consistent with lower appropriability of digital-investment returns.

Figure 6-4 visualizes these results in a dual-panel format: the left panel shows absolute DID coefficients with 95% confidence intervals; the right panel expresses effects as a percentage of the control mean, facilitating direct comparison of relative magnitudes

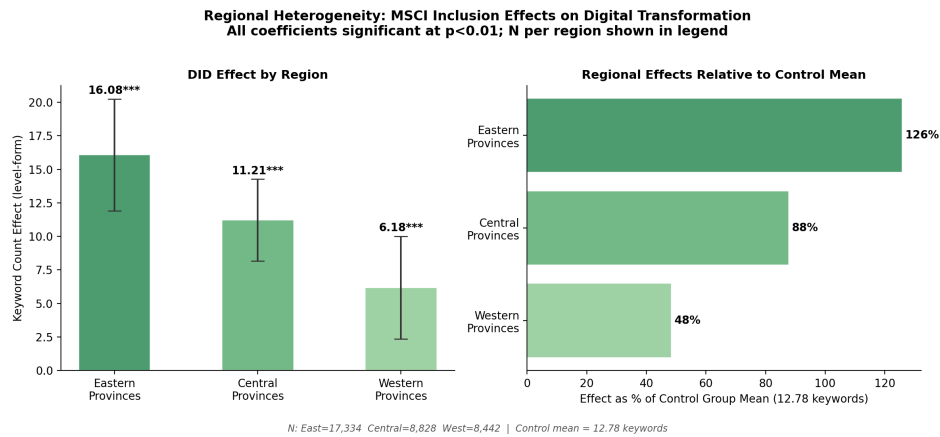


Figure 6-4 Regional Heterogeneity in MSCI Inclusion Effects

across regions.

6.4.2 Heterogeneity Analysis Based on Industry Level

Panel B of Table 6-8 reports the industry-level heterogeneity estimates. Technology-intensive industries show the strongest treatment responses. Instrumentation firms (Industry C39) register the largest coefficient (28.130***), followed by electrical equipment manufacturers (Industry C38: 13.032***) and transportation equipment (Industry C27: 2.764***). The high-tech versus low-tech differential is 50.589** ($p = 0.035$), which is statistically significant and economically large. The estimate is roughly four times the control mean.

6.4.3 Heterogeneity Analysis Based on Technology Domain

Table 6-9 decomposes heterogeneity at three levels to avoid mixing structurally different margins. Panel A isolates broad sector differences, Panel B captures regional moderation, and Panel C tests within-manufacturing technology-depth gradients (C1/C2/C3). Reading the table in this order helps separate composition effects from true moderation effects.

Panel C of Table 6-9 presents effects by manufacturing technology subcategory. High-end manufacturing (C3) shows the strongest effect (12.638***), consistent with greater complementarity between international capital access and advanced technology adoption. Basic manufacturing (C2: 0.930, $p = 0.117$) and equipment manufacturing (C1: 1.092, $p = 0.333$) show weaker, insignificant responses.

Table 6-9 Technology Intensity and Regional Heterogeneity Summary

Panel A: Broad Industry Groups				
Variable	(1) Manufacturing	(2) Info Services	(3) Finance	(4) Utilities
MSCI $\times$ Post	7.318*** (0.762)	−8.007 (7.311)	−4.056 (4.185)	0.431 (0.664)
Firm-FE	Yes	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
<i>N</i>	24,032	2,520	1,771	1,125
Panel B: Regional Moderation				
Variable	(5) East	(6) Central	(7) West	(8) Regional Int.
Coefficient of Interest	16.084*** (2.125)	11.210*** (1.559)	6.185*** (1.955)	10.446*** (2.403)
Firm-FE	Yes	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
<i>N</i>	17,334	8,828	8,442	34,604
Panel C: Manufacturing Technology Subcategories				
Variable	(9) C3	(10) C2	(11) C1	
MSCI $\times$ Post	12.638*** (1.168)	0.930 (0.592)	1.092 (1.129)	
Firm-FE	Yes	Yes	Yes	
Year-FE	Yes	Yes	Yes	
Controls	Yes	Yes	Yes	
<i>N</i>	14,019	6,952	2,322	

Notes: Entries report coefficients with standard errors in parentheses. Panel A joint test: $F = 97.26$ ($p < 0.001$). Panel C cross-category test: $F = 147.21$ ($p < 0.001$). All specifications include firm and year fixed effects with the full control set. Pre-treatment control mean is 12.78 keywords. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Panels A and B provide additional structure to this interpretation. In Panel A, the positive manufacturing coefficient is not replicated uniformly across all sectors, which confirms that pooled estimates can hide substantial sectoral dispersion. In Panel B, the east-central-west gradient persists alongside a significant regional interaction term, indicating that local institutional and capability conditions systematically moderate treatment conversion into digital outcomes. Together, Panels A–C support H3 as a moderated-response mechanism rather than a single homogeneous treatment effect.

6.4.4 Heterogeneity Analysis Based on Firm Characteristics

Across firm-characteristic dimensions, MSCI effects are stronger for: (i) financially constrained firms (higher KZ index at baseline), consistent with larger marginal value of international capital access; (ii) firms with weaker pre-treatment governance (lower independent director ratio), reflecting greater room for governance improvement; and (iii) firms with lower pre-treatment digital transformation scores, consistent with

convergence dynamics. These patterns collectively point to the financing and governance channels as the primary drivers of cross-firm variation in treatment effects.

Table 6-10 Evidence Mapping: Firm-Characteristic Heterogeneity

Firm-Characteristic Dimension	Pattern Reported	Evidence Source in Chapter 6	Interpretive Meaning
Financial constraints (higher baseline KZ)	Stronger effects	Financing-channel signals in Table 6-11 (-0.0318^{***} Bank Leverage; $+0.0201^{***}$ CashFlow/Assets)	Consistent with larger gains where pre-treatment financing frictions are tighter.
Pre-treatment governance quality (lower indep. director ratio)	Stronger effects	Governance-channel signals in Table 6-12 ($+1.8238^{***}$ Indep. Dir. Ratio; -0.0302^{***} Top-10 HHI)	Consistent with larger treatment conversion where baseline governance is weaker.
Pre-treatment digital transformation level (lower baseline DT)	Stronger effects	Directional subgroup pattern described in Section 6.4.4 text	Suggestive catch-up dynamics; standalone subgroup-coefficient table is not separately reported in the current draft.

Notes: This table maps the interpretation in Section 6.4.4 to supporting evidence already reported elsewhere in Chapter 6. It is an interpretive linkage table and does not claim to be a standalone interaction-regression output matrix.

6.5 Mechanism Evidence

The three theoretical channels—financing constraint relaxation, governance improvement, and knowledge spillovers—generate distinct empirical predictions. This study directly tests the first two channels by estimating Equation (4-1) with channel-specific proxy outcomes. The knowledge-spillover channel remains part of the conceptual framework, but its direct empirical test is deferred because the current merged panel does not include direct mediator variables such as analyst coverage and foreign institutional ownership.

Figure 6-5 illustrates the conceptual multi-channel pathway from $MSCI \times Post$ treatment to digital transformation.

Figure 6-6 presents the quantitative DID estimates for the six directly tested proxies in the financing and governance channels.

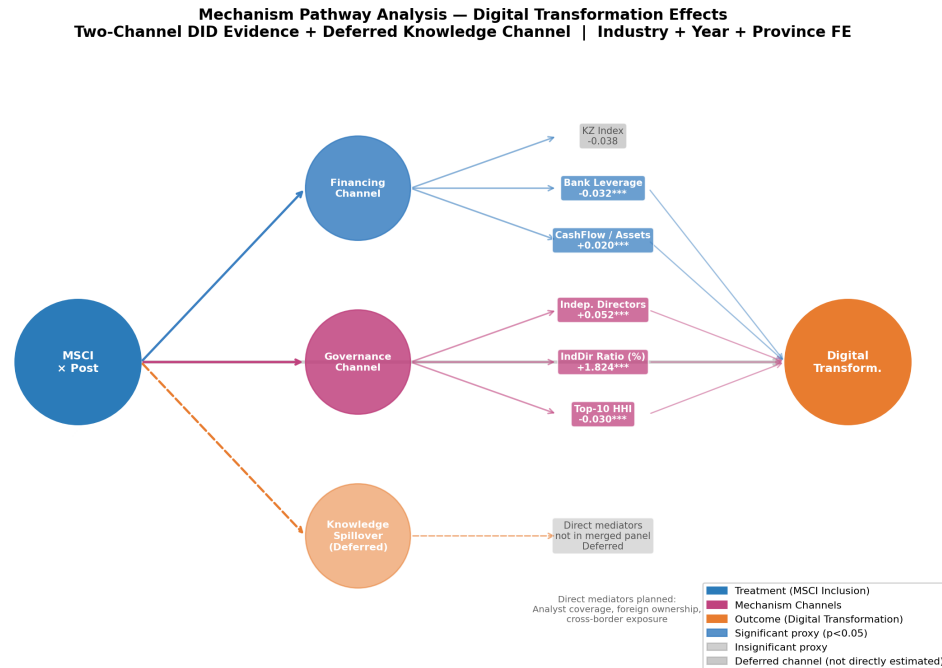


Figure 6-5 Mechanism Pathway: MSCI Inclusion to Digital Transformation

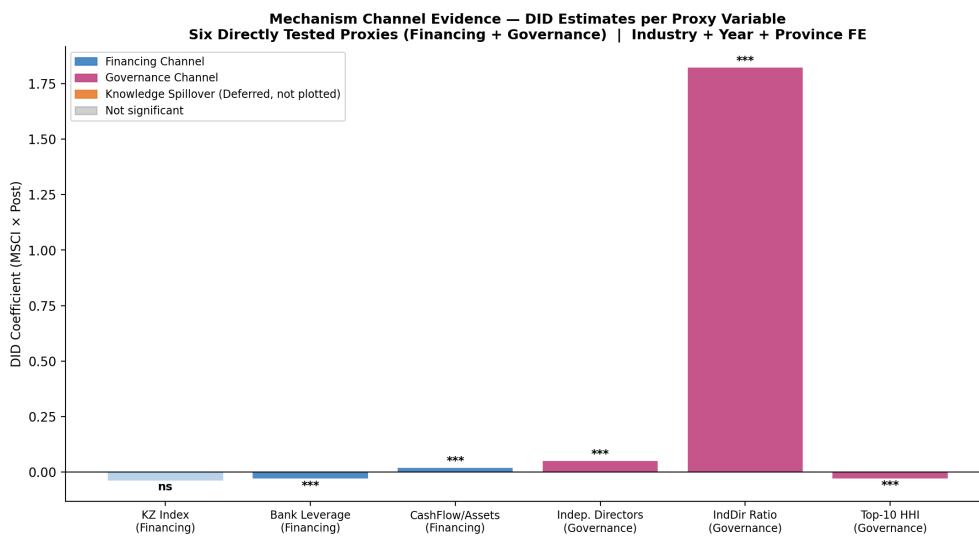


Figure 6-6 Mechanism Channel Evidence: DID Estimates for Financing and Governance Proxies

6.5.1 Financing Constraint Channel

The financing-constraint channel predicts that MSCI inclusion lowers the effective cost of external capital for treated firms.

To make this channel operational, the study uses three complementary firm-year proxies with distinct measurement logic. The KZ index is treated as a composite financing-friction score where larger values indicate tighter constraints. Bank leverage is the bank-debt reliance ratio in the merged panel (scaled by firm size), so lower

values indicate reduced dependence on bank credit after treatment. CashFlow/Assets is operating cash flow scaled by total assets and captures internally generated liquidity available for digital investment. All three mechanism regressions retain the same identification structure as the baseline model (Industry+Year+Province fixed effects with firm-clustered standard errors), so channel coefficients remain directly comparable to the main DID specification.

Table 6-11 Mechanism Analysis: Financing Channel

Variable	(1) KZ Index	(2) Bank Leverage	(3) CashFlow/Assets
MSCI $\times$ Post	-0.0385 (0.0733)	-0.0318*** (0.0027)	0.0201*** (0.0035)
Controls	Yes	Yes	Yes
Industry-FE	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes
Province-FE	Yes	Yes	Yes
<i>N</i>	28,339	23,552	34,770

Notes: Entries report MSCI $\times$ Post coefficients with firm-clustered standard errors in parentheses. *N* varies by proxy due to data availability in CSMAR. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The financing evidence is mixed and should be interpreted cautiously. The KZ coefficient is negative (-0.0385) but imprecise ($SE = 0.0733$, $p = 0.599$), so the composite financing-constraint index does not move significantly in this sample. In contrast, Bank Leverage declines significantly (-0.0318^{***} , $p < 0.001$), equivalent to about a 3.2 percentage-point decline in bank-debt reliance, and CashFlow/Assets increases by 0.0201^{***} ($p < 0.001$), roughly a 2.0 percentage-point rise in internal liquidity.

Taken together, this pattern is more consistent with financing-structure adjustment and improved financing flexibility than with a broad uniform relaxation of underlying financing frictions. Lower bank leverage can reflect debt-equity rebalancing, and higher cash flow may partly capture jointly evolving operating performance. Therefore, the financing channel is supported directionally but interpreted as moderate rather than definitive.

6.5.2 Governance Channel

International institutional investors are documented advocates of governance reform as a condition of sustained capital allocation^[73].

This channel is measured using board-structure and ownership-concentration variables that map directly to monitoring intensity. $N(\text{Independent Directors})$ is the count of independent directors and captures board monitoring capacity in levels. Independent Director Ratio (%) scales independent-director seats by total board size, so it captures compositional independence rather than board expansion alone. Top-10 Ownership HHI is a Herfindahl-type concentration index over top-shareholder holdings; larger values indicate tighter control concentration, while smaller values indicate more dispersed monitoring rights. Estimation again uses the same DID framework and fixed-effect structure as the baseline model, ensuring interpretation consistency across channels.

Table 6-12 Mechanism Analysis: Governance Channel

Variable	(1) Indep. Dir. Count	(2) Indep. Dir. Ratio	(3) Top-10 HHI
MSCI $\times$ Post	0.0518*** (0.0152)	1.8238*** (0.5627)	−0.0302*** (0.0054)
Controls	Yes	Yes	Yes
Industry-FE	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes
Province-FE	Yes	Yes	Yes
N	26,522	20,136	34,753

Notes: Entries report MSCI $\times$ Post coefficients with firm-clustered standard errors in parentheses. The HHI coefficient is expected to be negative if governance improves through ownership dispersion. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Governance estimates are directionally consistent and statistically significant, but interpretation requires institutional caution. MSCI-included firms add independent directors (+0.0518***, $p = 0.0007$), and the independent-director ratio rises by 1.8238 percentage points ($p = 0.0012$). Top-10 Ownership HHI declines (−0.0302***, $p < 0.001$), indicating more dispersed ownership.

These patterns are consistent with stronger external monitoring and disclosure discipline, yet they do not by themselves prove deep internal governance transformation. In the Chinese setting, independent-director ratios are partly compliance-driven, and ownership dispersion can also reflect passive benchmark-tracking flows after index inclusion. Accordingly, the governance channel is interpreted as evidence of strengthened external oversight conditions rather than definitive governance optimization.

6.5.3 Knowledge Spillover Channel

The knowledge-spillover mechanism remains theoretically central, but the current sample provides limited direct empirical support because key mediators are not yet merged into the estimation panel. Recent evidence confirms that digital transformation can generate information spillovers across firms and market participants^[57], and that spillover intensity may vary with strategic competition conditions in manufacturing settings^[58]. A direct test in this setting requires mediator variables that map to the chain MSCI inclusion → external information intermediaries/investors → firm learning and implementation. In this context, analyst coverage (and optionally analyst forecast dispersion) and foreign institutional ownership are the preferred first-stage mediators. Because these variables are not yet integrated into the current merged panel, this study does not report a direct knowledge-channel coefficient and avoids proxy substitution that could blur causal interpretation.

For transparency, the intended measurement protocol is as follows: analyst coverage is defined as the number of analysts issuing firm-year forecasts; analyst forecast dispersion is the cross-analyst standard deviation of earnings forecasts; foreign institutional ownership is the fraction of shares held by foreign institutions in each firm-year; and cross-border exposure proxies include export intensity, overseas subsidiary presence, and foreign director participation. These variables are conceptually close to the mechanism chain and therefore preferred over indirect substitutes. As a supplementary check, Appendix Table 0-1 reports an exploratory peer-spillover proxy regression based on leave-one-out industry-year peer digital transformation. The interaction term (MSCI × PeerDT) is positive but statistically insignificant ($p = 0.114$), so it is reported only as directional context rather than direct mechanism evidence. The deferred design choice is methodological rather than substantive: the study prioritizes identification quality by reporting no direct-mediator coefficient instead of reporting a weakly interpretable proxy-based estimate.

To maintain analytical transparency for this deferred channel, Table 6-13 reports a structured measurement and identification roadmap rather than a placeholder pseudo-regression.

Table 6-13 Knowledge Spillover Channel: Direct-Test Data and Identification Requirements

Mediator (Preferred)	Mechanism Mapping	Expected Sign	Candidate Data Source	Current Status	Planned Empirical Test
Analyst coverage (and optionally analyst forecast dispersion)	MSCI inclusion → external information production → firm learning and implementation	+ (coverage), – (dispersion)	CSMAR analyst module / CNRDS analyst files	Not yet merged	DID and event-study on mediator; mediation-consistency test with digital-transformation outcomes
Foreign institutional ownership share	MSCI inclusion → foreign monitoring and benchmarking pressure → managerial upgrading	+	CSMAR ownership module / WIND foreign holding data	Not yet merged	DID on ownership response; interaction and mechanism regressions with firm and year FE
Cross-border exposure proxies (exports, overseas subsidiaries, foreign directors)	MSCI inclusion → global network exposure → knowledge diffusion	+	CSMAR segment/disclosure modules; annual-report hand-collected fields	Partially available, not harmonized	Supplementary mechanism test after variable harmonization and missing-data diagnostics

Notes: This table specifies the mediator architecture for the knowledge-spillover channel, including variable definitions, expected effect directions, candidate data sources, and planned empirical tests. Presenting the design at this stage preserves causal coherence while direct mediator series are being harmonized in the merged panel.

The row-by-row design in Table 6-13 clarifies what is missing and why. Analyst variables identify the information-production margin, foreign ownership identifies monitoring and benchmarking pressure, and cross-border exposure identifies network-based diffusion. Reporting these rows explicitly improves interpretability relative to an “NA” coefficient table because it links each mediator to a concrete data source, expected sign, and planned test design.

Overall, Section 6.5 provides direct mechanism evidence for financing and governance channels. For the knowledge-spillover channel, the study provides a transparent identification roadmap and data requirements, and reserves coefficient reporting for future work once direct mediators are fully integrated.

Cross-Channel Summary To synthesize mechanism evidence without conflating unlike units, Table 6-14 reports sign, precision, and sample coverage across channels under a common fixed-effect specification.

Table 6-14 demonstrates a clear ranking in empirical strength. In financing, two of

Table 6-14 Mechanism Analysis: Cross-Channel Summary

Channel	Variable	Coeff.	SE	p-val	N
Financing	KZ Index	−0.0385	0.0733	0.5992	28,339
Financing	Bank Leverage	−0.0318***	0.0027	0.000	23,552
Financing	CashFlow / Assets	0.0201***	0.0035	0.000	34,770
Governance	N(Indep. Directors)	0.0518***	0.0152	0.0007	26,522
Governance	Indep. Director Ratio (%)	1.8238***	0.5627	0.0012	20,136
Governance	Top-10 Ownership HHI	−0.0302***	0.0054	0.000	34,753
Knowledge (Deferred)	Direct mediators pending integration	—	—	—	—

Notes: All reported coefficients come from regressions with Industry+Year+Province FE and firm-clustered standard errors. Across the six directly tested proxies, five are significant at the 1% level and all are directionally consistent with theory. The deferred knowledge row is a status indicator, not a regression estimate.

three proxies are significant and suggest improved funding flexibility, while the composite KZ index remains imprecise. In governance, all three proxies are significant and theory-consistent, with interpretation centered on enhanced external monitoring conditions rather than guaranteed internal governance deepening. The deferred knowledge row is deliberately non-estimated and should be interpreted as a data-integration status marker rather than a null mechanism finding.

6.6 Discussion

This section steps back from individual coefficients and considers what the combined evidence implies for understanding how capital market liberalization shapes firm-level digital transformation.

6.6.1 Interpretation of Main Findings

The combined evidence supports a positive but estimator-sensitive relationship between MSCI inclusion and digital-transformation outcomes. TWFE models produce large positive coefficients, but cohort-based modern DID diagnostics are markedly smaller and often imprecise. This divergence is consistent with heterogeneous treatment effects and potential TWFE weighting distortions under staggered adoption.

Accordingly, the central interpretation is not that liberalization generates a uniform large effect for all treated firms. Instead, the data suggest moderate and uneven treatment responses that vary across cohorts, contexts, and post-treatment horizons. The event-study dynamics remain informative because post-treatment effects accumulate rather than appearing as a one-period spike, which is more compatible with gradual strategic

upgrading than pure announcement noise.

At the same time, the outcome measure is disclosure-based and should be interpreted as a hybrid indicator of strategic communication and substantive implementation. The robustness evidence shows directional consistency across alternative constructions, but cannot fully separate symbolic disclosure from realized operational transformation.

Mechanism evidence is similarly nuanced. Financing results are mixed and align more with financing-structure adjustment than broad constraint relief. Governance proxies are strong in sign and precision, but best read as evidence of stronger external monitoring and disclosure discipline, with compliance and passive-ownership channels still plausible. For knowledge spillovers, the study reports a transparent deferred-testing roadmap rather than over-interpreting proxy estimates.

6.6.2 Implications for the Capital Market Liberalization Literature

These findings extend the liberalization literature in a direction that has received limited attention. Most prior studies focus on financial outcomes—stock returns, liquidity, cost of capital—and treat real-side effects as secondary. The evidence here suggests that liberalization can influence how firms invest in technology and organizational capabilities, connecting financial integration to longer-term productivity-related behavior.

The mechanism evidence adds specificity to this contribution. Rather than treating liberalization as a single undifferentiated shock, the analysis reveals that its effects operate through at least two distinct pathways—governance strengthening and financing-structure adjustment—with different degrees of empirical precision. This distinction matters for policy because it implies that the real-economy benefits of financial opening depend partly on the governance and institutional conditions that accompany it.

The heterogeneity results reinforce this point. Treatment effects are strongest in eastern provinces, technology-intensive industries, and high-end manufacturing—settings where the complementary conditions for digital adoption are already in place. This pattern is consistent with an absorptive-capacity framework and suggests that liberalization amplifies existing capabilities rather than creating them from scratch.

6.6.3 Policy Implications

The results imply that capital-market liberalization can support digital upgrading, but only under clear complementary conditions. Treatment effects are stronger where absorptive capacity, institutional quality, and implementation readiness are already relatively high. Liberalization alone is therefore unlikely to generate uniform technology upgrading across firms.

For policymakers, this supports a conditional policy package: financial opening should be paired with governance reforms, digital infrastructure investment, innovation-financing support, and capability development in lagging regions and sectors. For firms, the evidence suggests that capital access is most useful when combined with organizational implementation capacity rather than disclosure expansion alone. For international investors, stewardship and monitoring can matter, but the strongest long-run gains are more likely when monitoring incentives translate into substantive, not merely symbolic, transformation commitments.

Chapter 7 synthesizes these empirical results into overall contributions, limitations, and future research priorities.

Chapter 7 Conclusion

This chapter synthesizes the main findings, contributions, and limitations of the study, and outlines directions for future research.

7.1 Summary of Main Findings

Using a difference-in-differences framework on 35,264 firm-year observations (2010–2022) and NLP-based digital-transformation measures, this study finds a positive but estimator-sensitive relationship between MSCI inclusion and firm digital-transformation outcomes. The main evidence can be summarized as follows:

1. TWFE benchmark estimates are positive and statistically strong (0.394 log points in the controlled specification), but cohort-based modern DID diagnostics are materially smaller and less precise, indicating substantial sensitivity to estimator choice under staggered treatment timing;
2. event-study and placebo evidence supports treatment timing plausibility, while alternative outcomes and disclosure-length checks support directional robustness under a hybrid disclosure-substantive measure interpretation;
3. heterogeneity patterns are strongest in high-technology and institutionally developed settings, consistent with absorptive-capacity and institutional-complementarity mechanisms;
4. mechanism evidence is mixed across channels: financing results align more with financing-structure adjustment than broad constraint relaxation, governance proxies suggest strengthened external monitoring conditions with institutional caveats, and direct knowledge-spillover testing remains deferred because key mediators are not yet fully integrated.

Overall, the findings suggest that capital-market liberalization can support digital upgrading, but the average effect is moderate, heterogeneous, and conditional on implementation capacity and institutional context. The study therefore emphasizes calibrated interpretation over universal causal claims.

7.2 Contributions

1. **Identification-oriented evidence on digital transformation drivers:** This study provides quasi-experimental evidence that international financial integration is associated with higher firm-level digital transformation intensity in an emerging-market context, while explicitly documenting estimator sensitivity and treatment heterogeneity under staggered adoption.
2. **Integration of text-as-data with econometric identification:** The study provides evidence that validated NLP-based disclosure measures can be embedded in rigorous DID workflows when accompanied by normalization, placebo testing, and alternative-outcome sensitivity checks.
3. **Unified mechanism architecture:** Instead of testing channels separately, the study evaluates financing and governance pathways under a common treatment design, generating internally coherent evidence across six directly tested proxy variables.
4. **Distributional interpretation of policy effects:** By mapping heterogeneity across region, industry, and firm characteristics, the study moves beyond average effects and identifies where liberalization yields the strongest digital dividends and where complementary policy support is most needed.

These contributions are mutually reinforcing: stronger identification supports mechanism interpretation, mechanism evidence improves policy relevance, and transparent measurement improves replicability and external credibility.

7.3 Limitations

Five limitations merit explicit acknowledgment.

First, the text-based outcome captures disclosed digital orientation and intensity rather than direct implementation records. Although normalization and robustness checks mitigate this concern, symbolic disclosure incentives cannot be fully separated from substantive transformation in the current measure.

Second, treatment-effect magnitudes are sensitive to estimator choice in staggered-treatment settings. Modern DID diagnostics indicate that conventional TWFE estimates

may overstate average effects when treatment responses are heterogeneous across cohorts and time.

Third, MSCI inclusion is rule-based but not randomly assigned. Treated firms differ systematically from untreated firms on pre-treatment characteristics such as size, liquidity, and governance, and residual bias from unobserved selection cannot be fully ruled out.

Fourth, mechanism measurement remains uneven across channels. Financing and governance proxies are observable in the current panel, but direct knowledge-spillover mediators (e.g., analyst coverage and foreign institutional ownership) are not yet fully harmonized.

Fifth, external validity is bounded by the Chinese institutional context. Financial structure, disclosure regulation, and digital-economy conditions differ across emerging markets, so cross-country generalization should be made cautiously.

7.4 Future Work

Future research can advance along five directions.

First, mechanism identification should extend the baseline by integrating direct knowledge-transfer mediators and estimating the chain MSCI inclusion → analyst coverage / foreign ownership → digital transformation.

Second, heterogeneity-robust DID estimators and imputation-based event-study approaches should be implemented as primary estimators when longer post-treatment windows and larger cohort samples become available.

Third, cross-country comparative designs exploiting analogous benchmark-inclusion episodes can clarify which findings are China-specific and which generalize across emerging markets.

Fourth, extending the sample beyond 2022 will allow testing whether estimated effects persist, plateau, or reverse as inclusion factors and global portfolio weights evolve.

Fifth, richer NLP methods such as supervised classification, contextual embeddings, and topic evolution models can improve construct validity by distinguishing strategic rhetoric from operational implementation in corporate digital narratives.

Taken together, the study positions capital market liberalization as a potentially important but conditional contributor to firm-level digital upgrading, with effects that are heterogeneous, estimator-sensitive, and channel-specific in transmission.

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Patience Fero

2025 年 5 月

Appendix

1 Exploratory Knowledge Spillover Proxy Test

To align the empirical design with recent spillover studies^[57-58] while preserving identification discipline, this appendix reports an exploratory proxy test based on currently available merged variables. The proxy is leave-one-out peer digital transformation within industry-year cells:

$$\text{PeerDT}_{i,t}^{(\text{ind},\text{loo})} = \frac{\sum_{j \in (\text{ind},t), j \neq i} \text{DT}_{j,t}}{N_{\text{ind},t} - 1} \quad (0-1)$$

and the exploratory interaction is $\text{MSCI}_{i,t} \times \text{PeerDT}_{i,t}^{(\text{ind},\text{loo})}$.

Table 0-1 Appendix: Exploratory Knowledge Spillover Proxy Regression

Variable	Coeff.	(SE)	p-val
MSCI (treatment exposure)	-2.0000	1.8336	0.2754
PeerDT (industry-year leave-one-out mean)	0.7046***	0.0689	0.0000
MSCI $\times$ PeerDT	0.2342	0.1481	0.1138
Observations	31,110		
Unique firms (cluster level)	3,921		
R^2	0.3556		

Notes: OLS with firm-clustered standard errors; controls include size, ROA, cashflow/assets, Tobin's Q , independent directors, and ownership HHI, with industry, province, and year fixed effects. The interaction coefficient is positive but statistically insignificant, so this table is interpreted as exploratory context rather than direct mechanism identification.

This appendix regression is reported as descriptive robustness context only. Because the interaction estimate is statistically insignificant and relies on a proxy rather than direct mediators, the result should be interpreted as directional background evidence for future mechanism integration.

